

Recap: Magnetic Fields of Various Current Distributions

The following table lists magnetic fields caused by several current distributions. In each case the conductor is carrying current I .

Current Distribution	Point in Magnetic Field	Magnetic Field Magnitude
Long, straight conductor	Distance r from conductor	$B = \frac{\mu_0 I}{2\pi r}$
Circular loop of radius a	On axis of loop	$B = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}}$
	At center of loop	$B = \frac{\mu_0 I}{2a}$ (for N loops, multiply these expressions by N)
Long cylindrical conductor of radius R	Inside conductor, $r < R$	$B = \frac{\mu_0 I}{2\pi} \frac{r}{R^2}$
	Outside conductor, $r > R$	$B = \frac{\mu_0 I}{2\pi r}$
Long, closely wound solenoid with n turns per unit length, near its midpoint	Inside solenoid, near center	$B = \mu_0 n I$
	Outside solenoid	$B \approx 0$
Tightly wound toroidal solenoid (toroid) with N turns	Within the space enclosed by the windings, distance r from symmetry axis	$B = \frac{\mu_0 N I}{2\pi r}$
	Outside the space enclosed by the windings	$B \approx 0$

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https://www.bilibili.com/video/BV1QZ4y1s74i/?spm_id_from=333.788.recommend_more_video.4

Recap: Vector Operation

Scalar Product (dot product)

Scalar product $\vec{A} \cdot \vec{B} = AB \cos \phi$

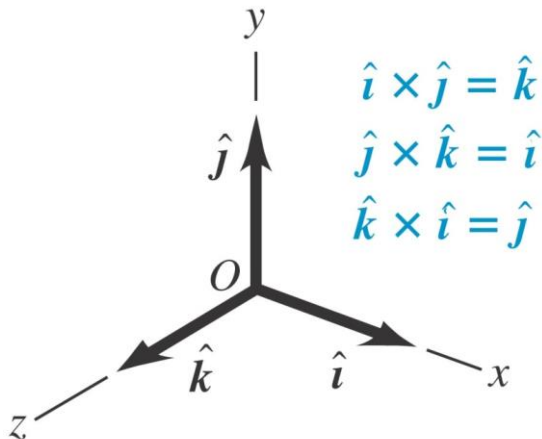


$$\oint_{\mathbf{A}=\partial V} \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\epsilon_0}$$

$$\oint_{\mathbf{A}=\partial V} \mathbf{B} \cdot d\mathbf{A} = 0$$

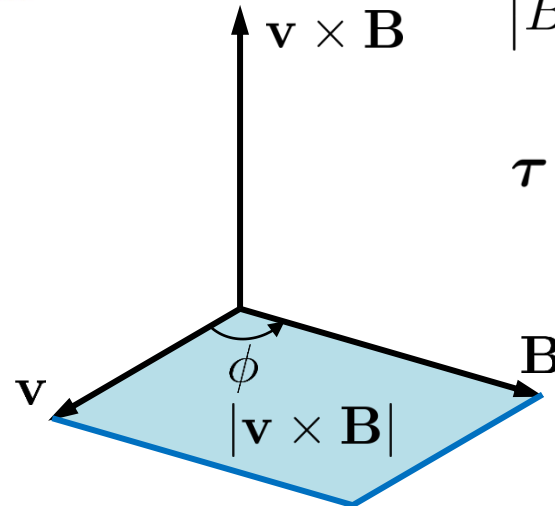
Vector Product (cross product)

A right-handed coordinate system



$$\mathbf{F}_B = q\mathbf{v} \times \mathbf{B} = q \begin{vmatrix} \mathbf{e}_x & \mathbf{e}_y & \mathbf{e}_z \\ v_x & v_y & v_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}$$



A brief review

Electrostatic

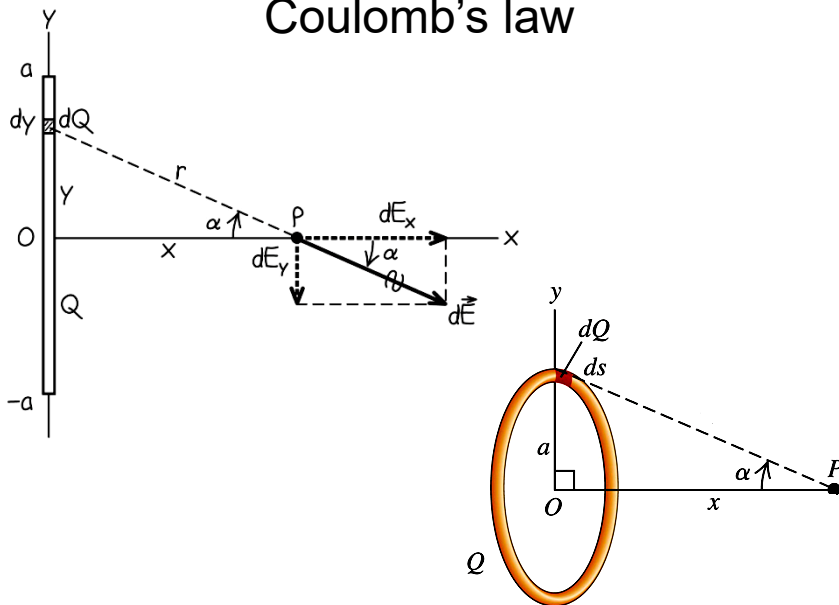
Force :

$$\vec{F} = q\vec{E}$$

Field :

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dQ}{r^2}$$



Coulomb's law

Magnetism

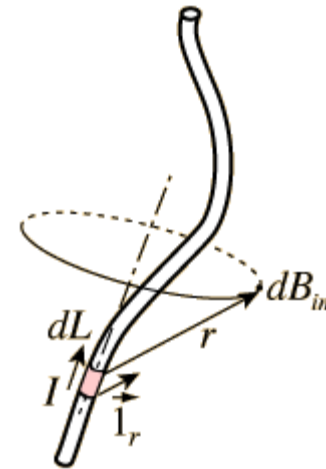
$$\vec{F} = q\vec{v} \times \vec{B}$$

$$\vec{F} = i\vec{l} \times \vec{B}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \hat{r}}{r^2}$$

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$$

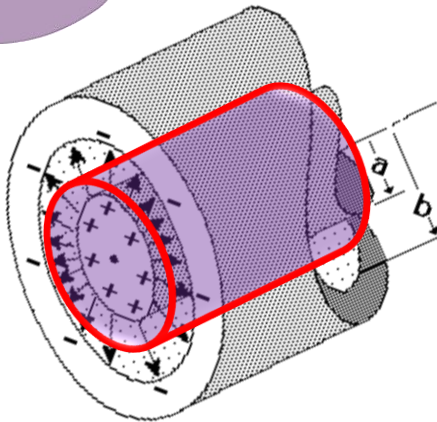
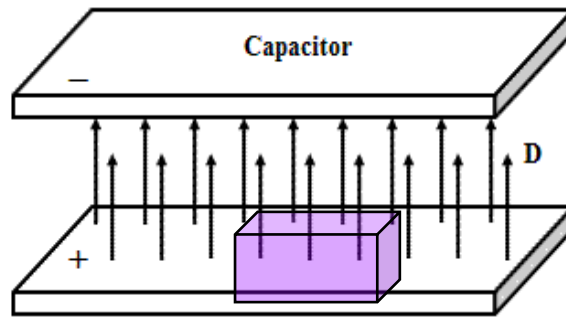
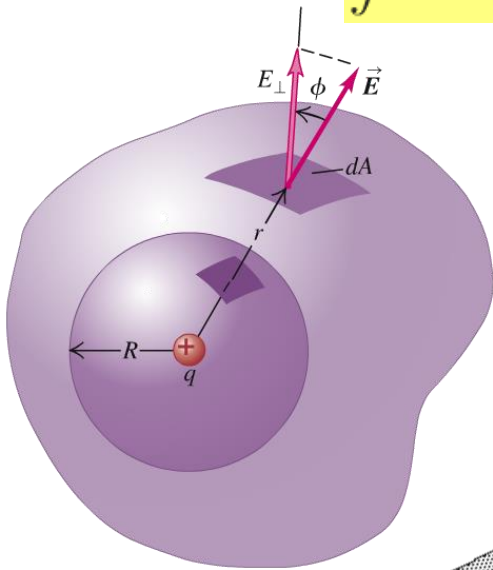
Biot-Savart Law



A brief review

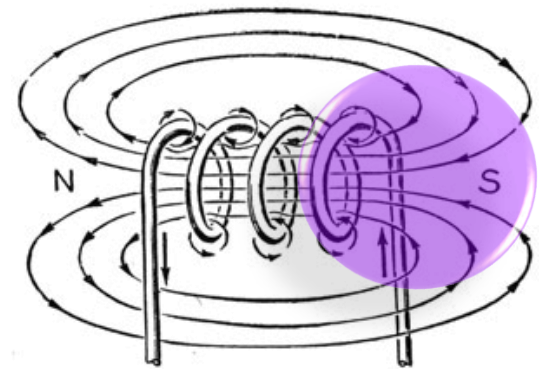
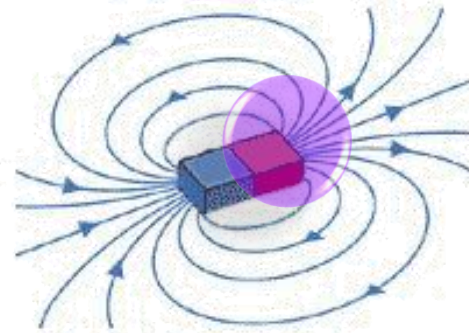
Gauss's law for $\vec{E}$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$



Gauss's law for $\vec{B}$

$$\oint \vec{B} \cdot d\vec{A} = 0$$

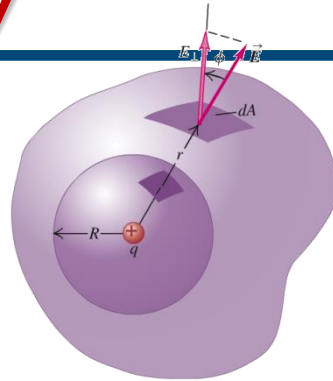


A brief review

Gauss's law for $\vec{E}$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$

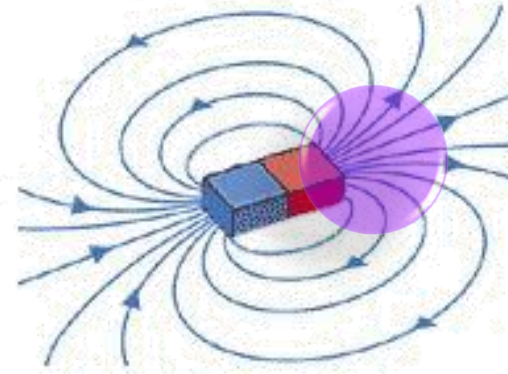
closed **surface**
→ **area**



Gauss's law for $\vec{B}$

$$\oint \vec{B} \cdot d\vec{A} = 0$$

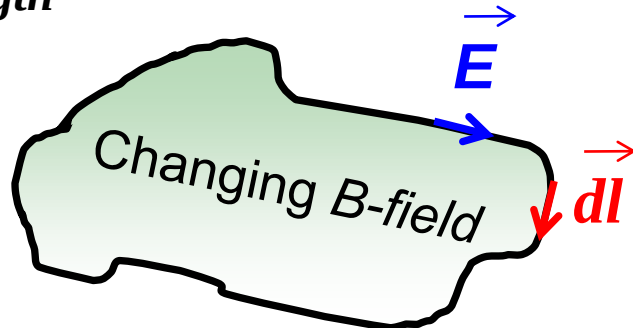
closed **surface**
→ **area**



Faraday's law

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{A}$$

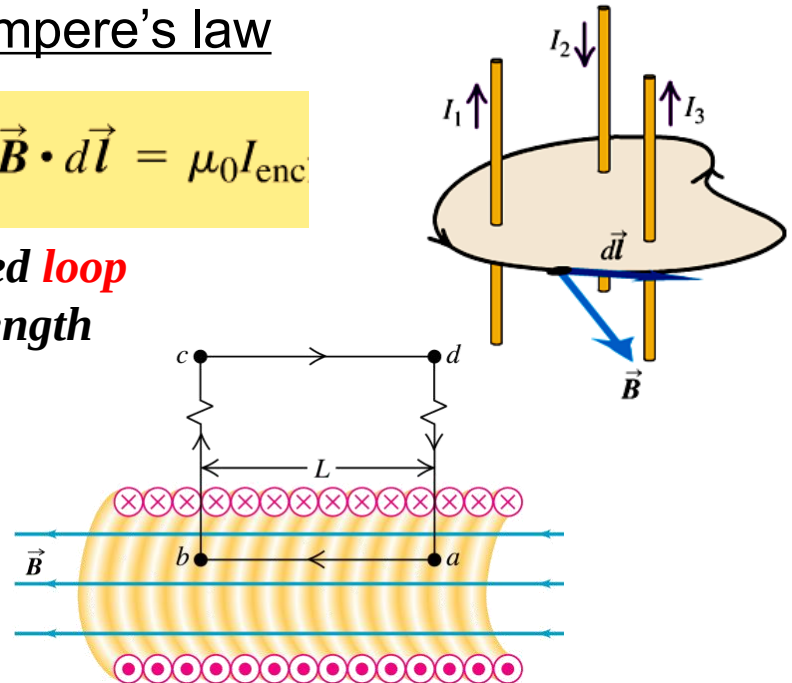
closed **loop**
→ **length**



Ampere's law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

closed **loop**
→ **length**



New Experiments: what to learn?



Ch30 Induction and Inductance

In the last lecture, we learned that a steady current could produce a steady magnetic field.

We have connected electricity to magnetism.

So, can we use a steady magnetic field to produce a steady current?

No, we cannot !

Instead, we found that a **changing magnetic field** can give **steady electric current**. And this is called electromagnetic induction.

Ch30 Key Points

1. Faraday's law and Lenz's law
2. Induction and energy transfers
3. Induced electric fields
4. Inductors and Inductance
5. Self-induction
6. *RL* circuits
7. Energy stored in a magnetic field
8. Energy density of a magnetic field
9. Mutual induction

The third Equation of the Maxwell's Equations

Differential Form

Integral Form

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \\ \nabla \cdot \mathbf{B} = 0 \end{array} \right.$$

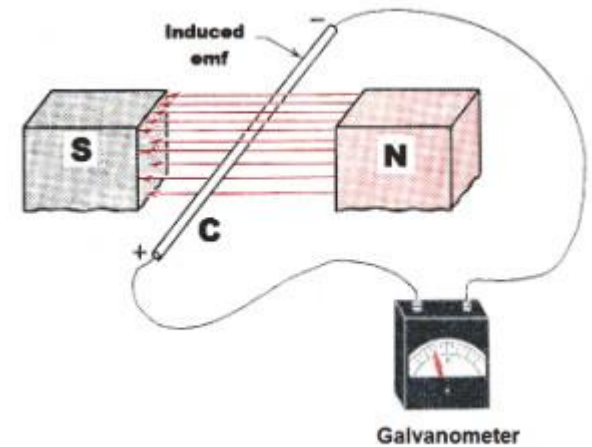
$$\left\{ \begin{array}{l} \oint_{\mathbf{A}=\partial V} \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\epsilon_0} \\ \oint_{\mathbf{A}=\partial V} \mathbf{B} \cdot d\mathbf{A} = 0 \end{array} \right.$$

$$\left\{ \begin{array}{l} \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \end{array} \right. \quad \left\{ \begin{array}{l} \oint_{\mathbf{l}=\partial \Sigma} \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int_{\Sigma} \mathbf{B} \cdot d\mathbf{\Sigma} \end{array} \right.$$

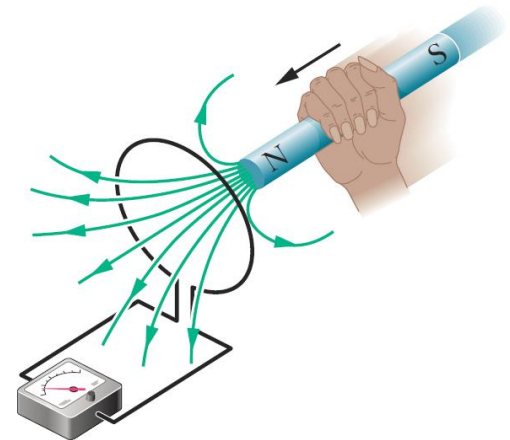
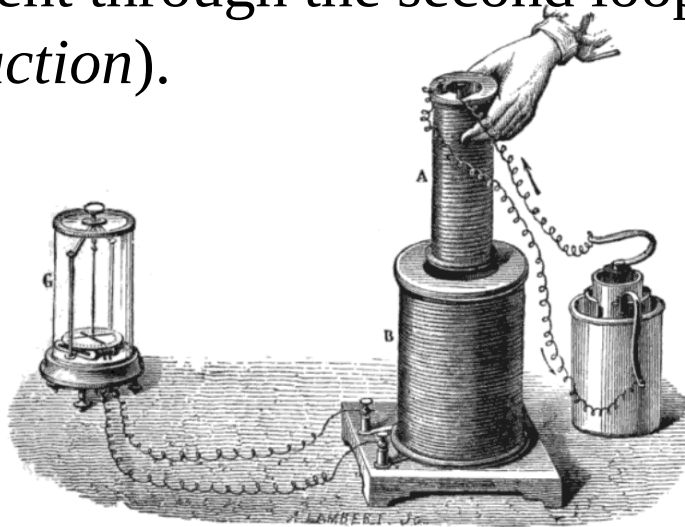
$$\left\{ \begin{array}{l} \nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \end{array} \right. \quad \left\{ \begin{array}{l} \oint_{\mathbf{l}=\partial \Sigma} \mathbf{B} \cdot d\mathbf{l} = \mu_0 \left(\int_{\Sigma} \mathbf{J} \cdot d\mathbf{\Sigma} + \epsilon_0 \frac{d}{dt} \int_{\Sigma} \mathbf{E} \cdot d\mathbf{\Sigma} \right) \end{array} \right.$$

Experiments by Michael Faraday

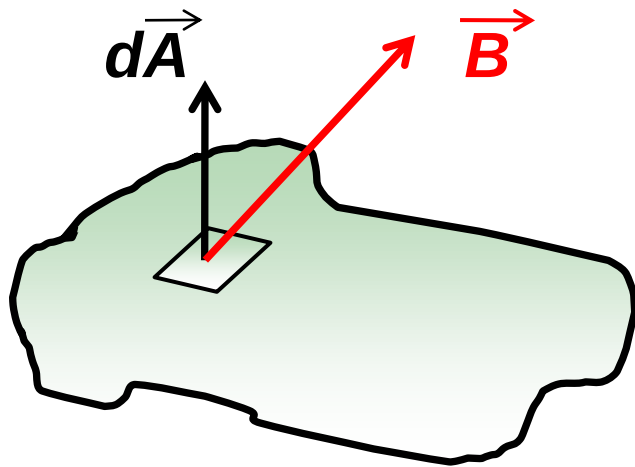
Experiment 1: Current flows through the coil when the position of the magnet changes with respect to the coil.



Experiment 2: Current flowing through the first loop generates a changing magnetic field, which in turn generates a current through the second loop (*mutual induction*).



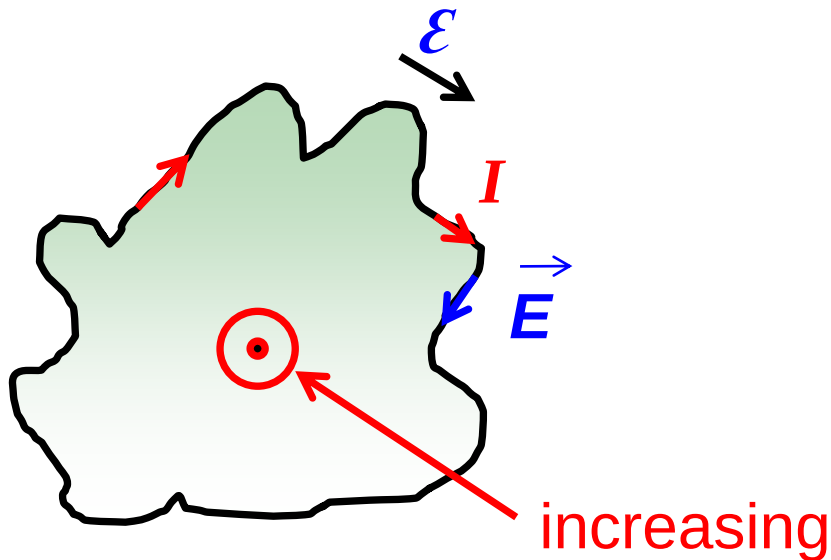
Magnetic flux Φ_B & Faraday's Law



From Chapter 28 we knew that the magnetic flux is defined as

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

In words: An *emf* is induced in the loop below, when the *magnetic flux* that passed through the loop is changing. The value of the induced *emf* and the induced current are determined by the *rate* at which the flux changes.

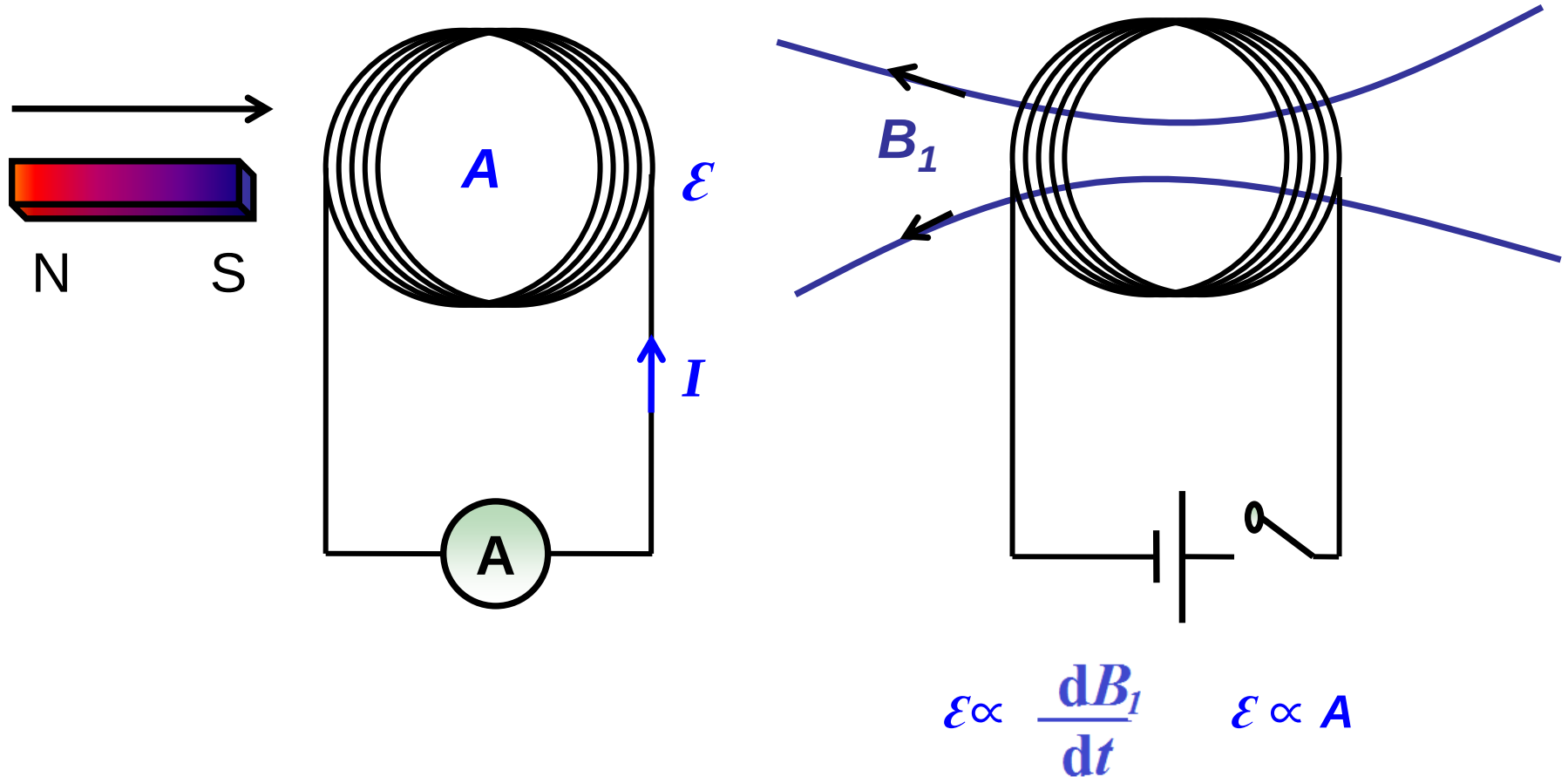


Faraday's Law

$$\begin{aligned} \mathcal{E} &= -\frac{d\Phi_B}{dt} = \frac{d}{dt} \int \vec{B} \cdot d\vec{A} \\ &= \oint \vec{E} \cdot d\vec{s} \end{aligned}$$

Faraday's induction experiment

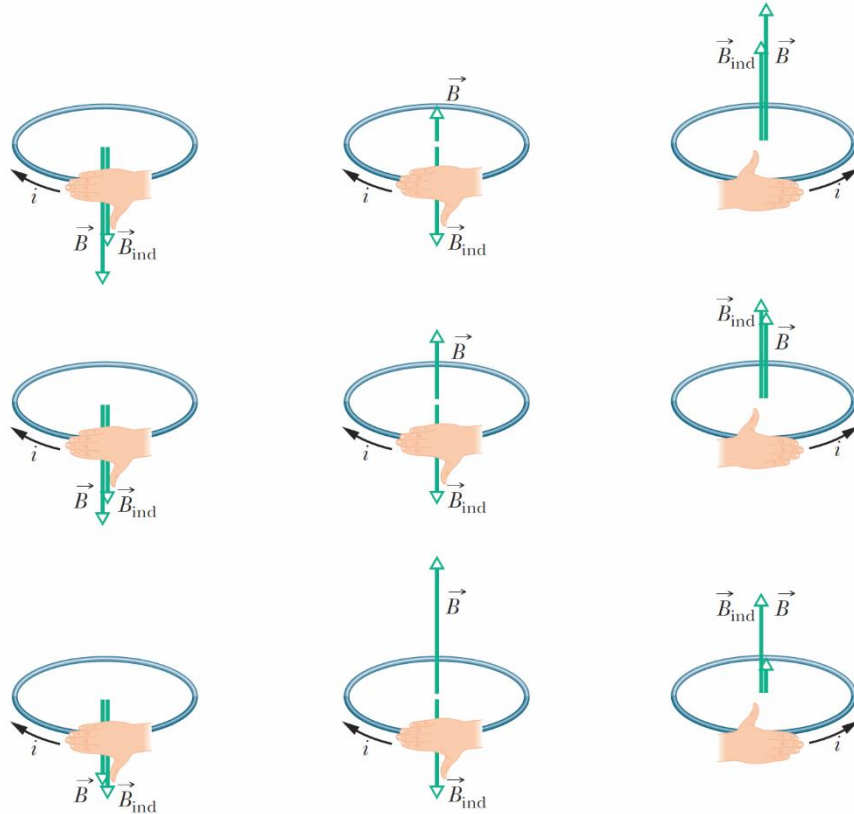
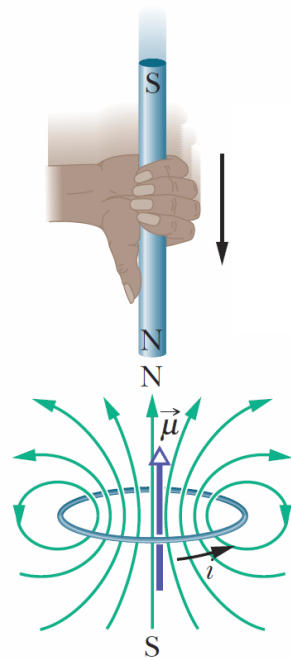
If there is a current, like we had with a battery, there must be a **emf** $\mathcal{E}$ and an ***E-field***



$\mathcal{E}$ is the result of the change of **magnetic flux** through the area A

Lenz's law

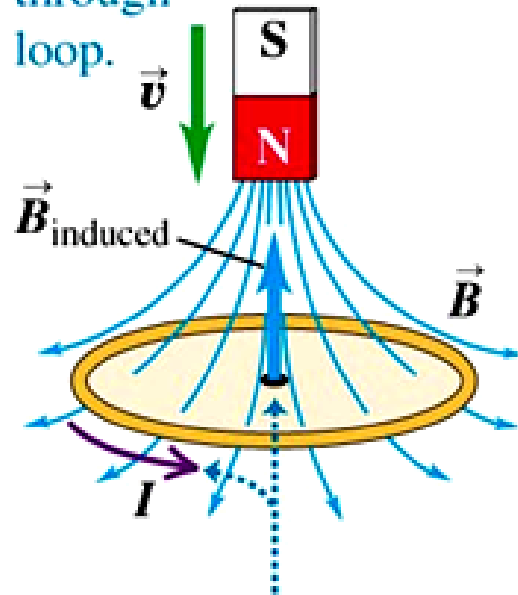
$$\mathcal{E} = - \frac{d\Phi_B}{dt}$$



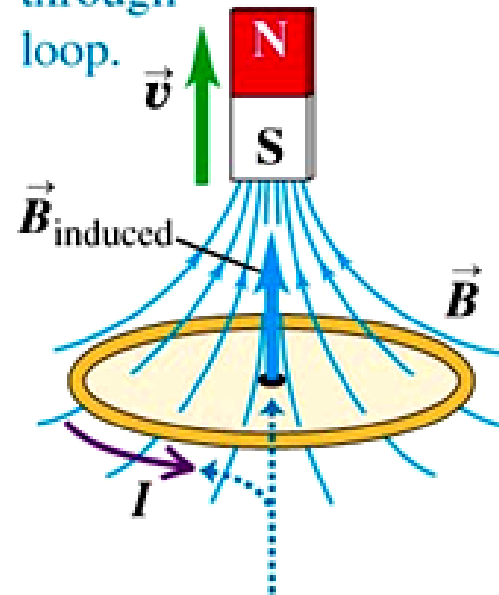
In words: An induced current has a direction such that the magnetic field *due to the current* opposes the *change* in the magnetic flux that induced the current.

Lenz's law

- (a) Motion of magnet causes *increasing downward flux* through loop.



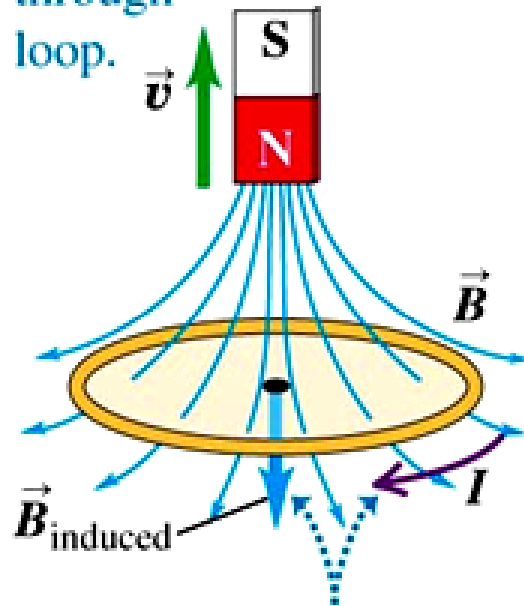
- (b) Motion of magnet causes *decreasing upward flux* through loop.



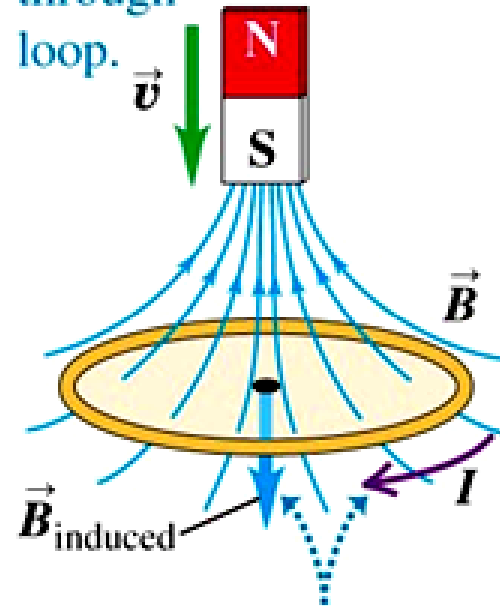
The induced magnetic field is *upward* to oppose the flux change. To produce this induced field, the induced current must be *counterclockwise* as seen from above the loop.

Lenz's law

- (c) Motion of magnet causes *decreasing downward flux* through loop.



- (d) Motion of magnet causes *increasing upward flux* through loop.

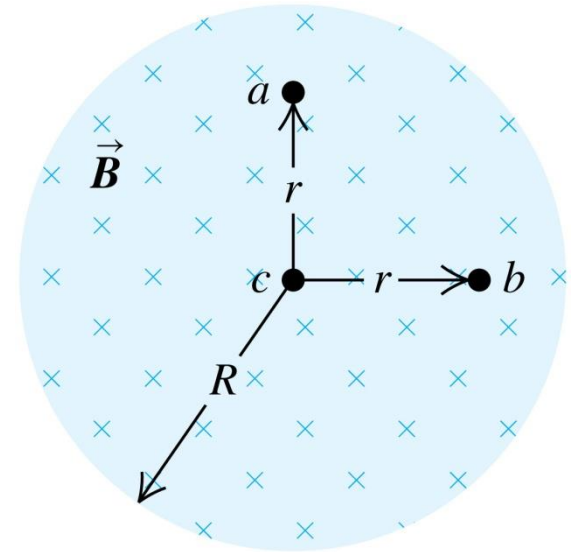


The induced magnetic field is *downward* to oppose the flux change. To produce this induced field, the induced current must be *clockwise* as seen from above the loop.

Question_Lenz's law

The drawing shows the uniform magnetic field inside a long, straight solenoid. The field is directed into the plane of the drawing and is increasing.

What is the direction of the *electric* force on a positive point charge placed at point *a*, *b*, and *c*?

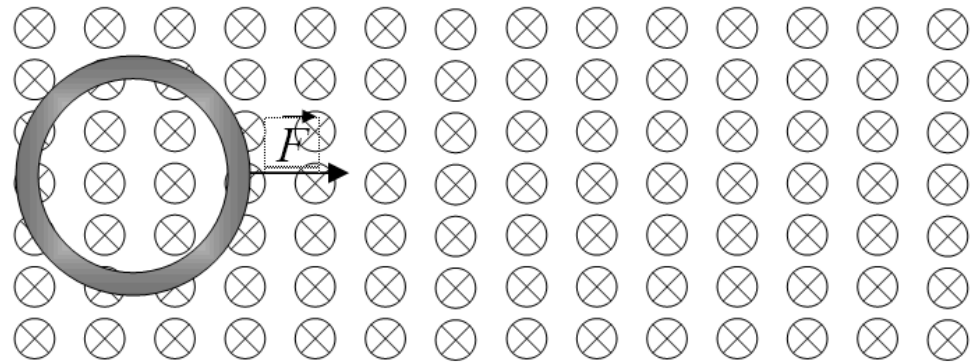


- A. to the left
- B. to the right
- C. straight up
- D. straight down
- E. misleading question—the electric force at this point is zero

Question_ Faraday's Law

A rigid, circular metal loop begins at rest in a uniform magnetic field directed away from you as shown. The loop is then pulled through the field toward the right, but does not exit the field. What is the direction of any induced current within the loop?

- a) clockwise
- b) counterclockwise
- c) No current is induced.

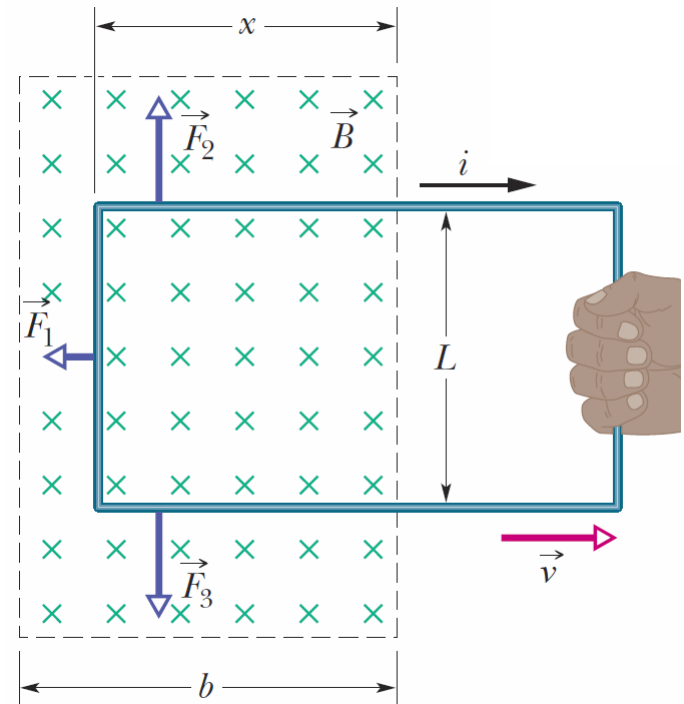


Question_ Faraday's Law: to change of loop area

When a loop is pulling out from the region of B -field, induction occurs since the magnetic flux *inside the loop* changes.

In the figure, if the loop is pulling to the right at a constant v , according to Faraday's law,

$$\begin{aligned} |\mathcal{E}_{\text{ind}}| &= \frac{d\Phi_B}{dt} = \frac{d}{dt} \int_{\mathbf{A}} \mathbf{B} \cdot d\mathbf{A} \\ &= \frac{d}{dt} (BA) = B \frac{dA}{dt} \\ &= BL \frac{dx}{dt} = BLv. \quad (\text{clockwise}) \end{aligned}$$



Induction and energy transfers

Therefore the induced current is

$$i = \frac{|\mathcal{E}_{\text{ind}}|}{R} = \frac{BLv}{R}. \quad (\text{clockwise})$$

According to Chapter 29, this current generates a force

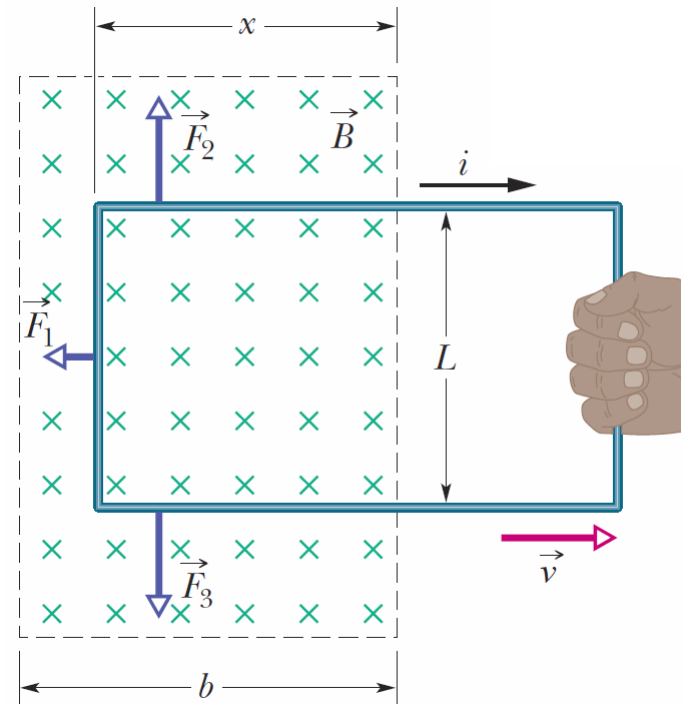
$$F = iLB = \frac{B^2 L^2 v}{R}.$$

Since v is constant, the magnetic force generates power

$$P = Fv = \frac{B^2 L^2 v^2}{R},$$

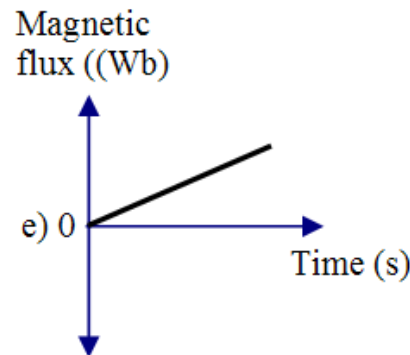
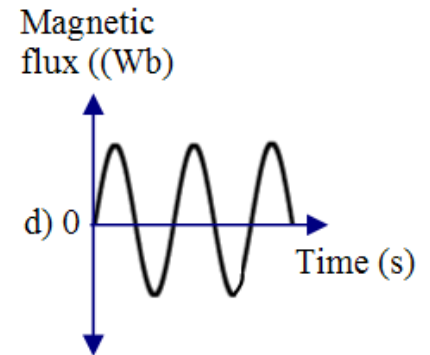
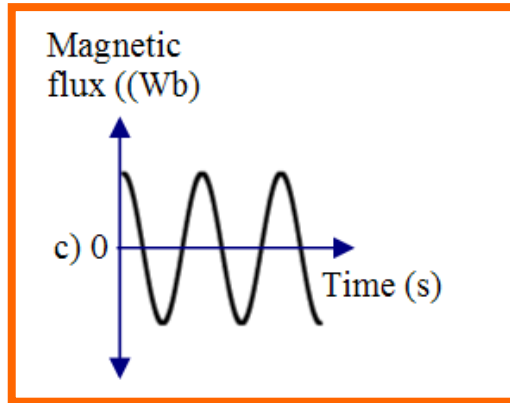
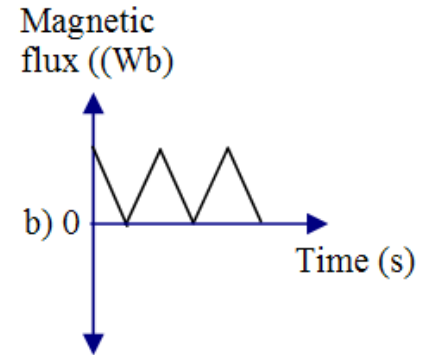
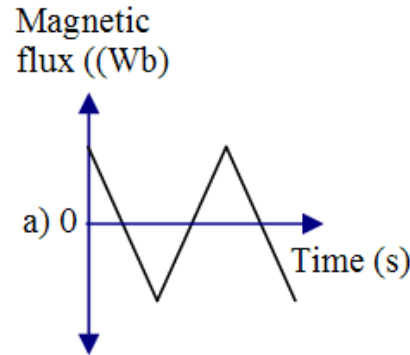
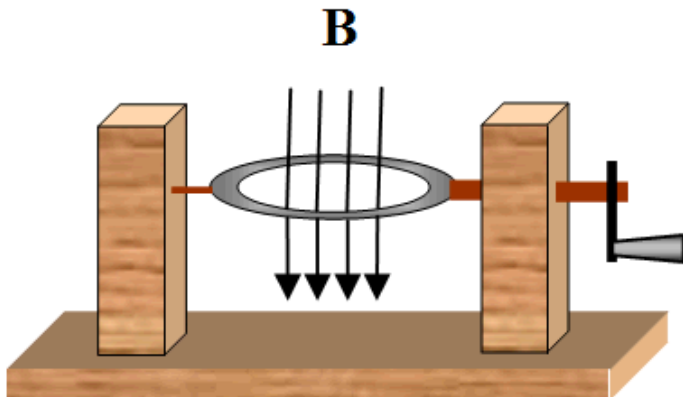
which equals to the thermal power

of the current loop:
$$P_{\text{thermal}} = i^2 R = \left(\frac{BLv}{R} \right)^2 R = \frac{B^2 L^2 v^2}{R}.$$



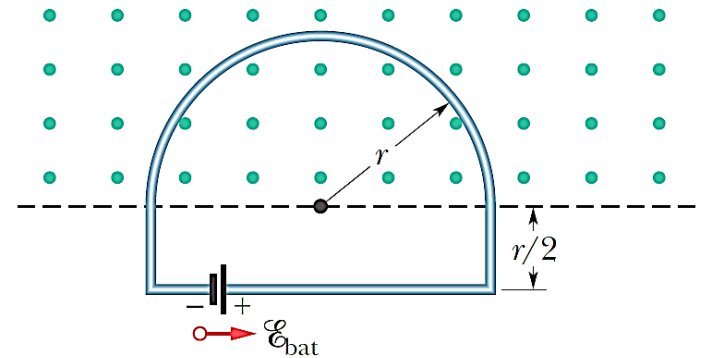
Question_Faraday's Law: to change of loop area

A circular ring is rotated clockwise at a constant rate for an extended period of time using the apparatus shown. Which of the graphs below correctly shows the magnetic flux through the ring as a function of time? Note: At time $t = 0$ s, the plane of the ring is perpendicular to the direction of the magnetic field.



Question_Faraday's Law: to change of magnetic field

A conducting loop consisting of a half-circle of radius $r = 0.20$ m and three straight sections. The half circle lies in a uniform magnetic field that is directed out of the page; the field magnitude is given by $B = 4.0t^2 + 2.0t + 3.0$, with B in Teslas and t in seconds. An ideal battery with emf $\mathcal{E}_{\text{bat}} = 2.0$ V is connected to the loop. The resistance of the loop is 2.0Ω .



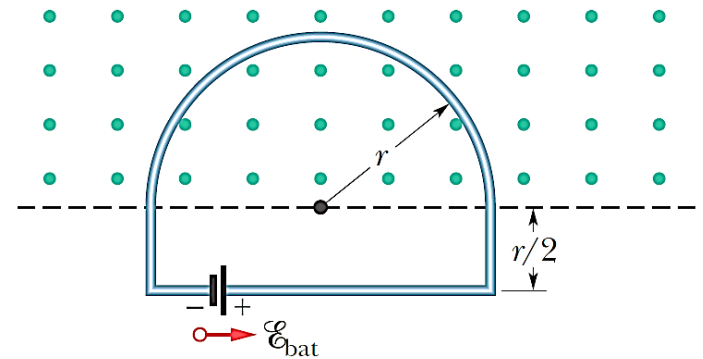
Question_Faraday's Law: to change of magnetic field

(a) What are the magnitude and direction of the emf $\mathcal{E}_{\text{ind}}$ induced around the loop by the field B at $t = 10$ s?

Solve: Since B is increasing with t , the field generated by $\mathcal{E}_{\text{ind}}$ would be pointing inside the page, so $\mathcal{E}_{\text{ind}}$ is in the **clockwise** direction.

Magnitude:

$$\begin{aligned} |\mathcal{E}_{\text{ind}}| &= \frac{d\Phi_B}{dt} \\ &= \frac{d}{dt} \int_{\mathbf{A}} \mathbf{B} \cdot d\mathbf{A} = \frac{d}{dt} (BA) = \frac{1}{2} \pi r^2 \frac{d}{dt} (4t^2 + 2t + 3) \\ &= \pi r^2 (4t + 1) = \pi (0.2 \text{ m})^2 (4 \times 10 \text{ s} + 1) = 5.152 \text{ V}. \end{aligned}$$



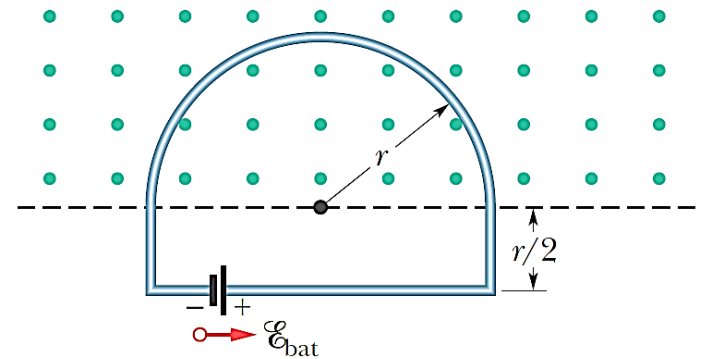
Question_Faraday's Law: to change of magnetic field

(b) What is the current in the loop at $t = 10$ s?

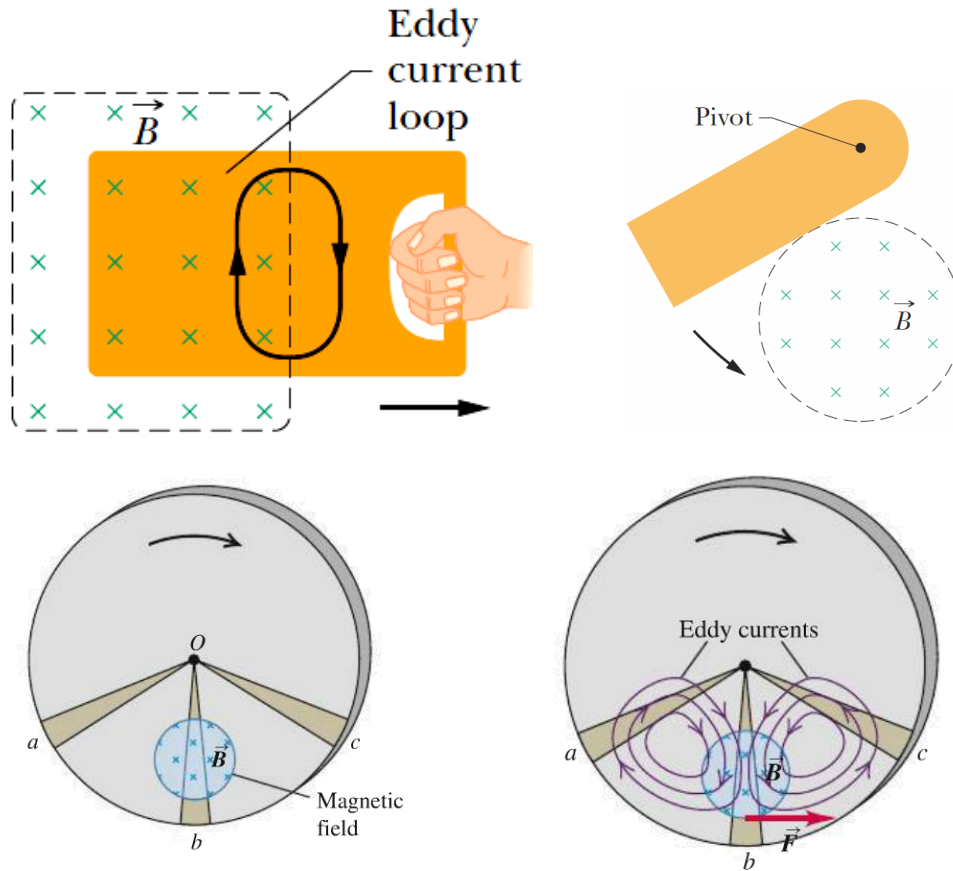
Solve: Since $\mathcal{E}_{\text{ind}}$ is clockwise and $\mathcal{E}_{\text{bat}}$ is counterclockwise, the magnitude of the net emf is the difference of the two, pointing clockwise (since $\mathcal{E}_{\text{ind}}$ is larger).

So

$$\begin{aligned} i &= \frac{\mathcal{E}_{\text{net}}}{R} = \frac{\mathcal{E}_{\text{ind}} - \mathcal{E}_{\text{bat}}}{R} \\ &= \frac{5.152 \text{ V} - 2.0 \text{ V}}{2.0 \text{ } \Omega} = 1.58 \text{ A.} \quad (\text{clockwise}) \end{aligned}$$



Application: *Eddy current*

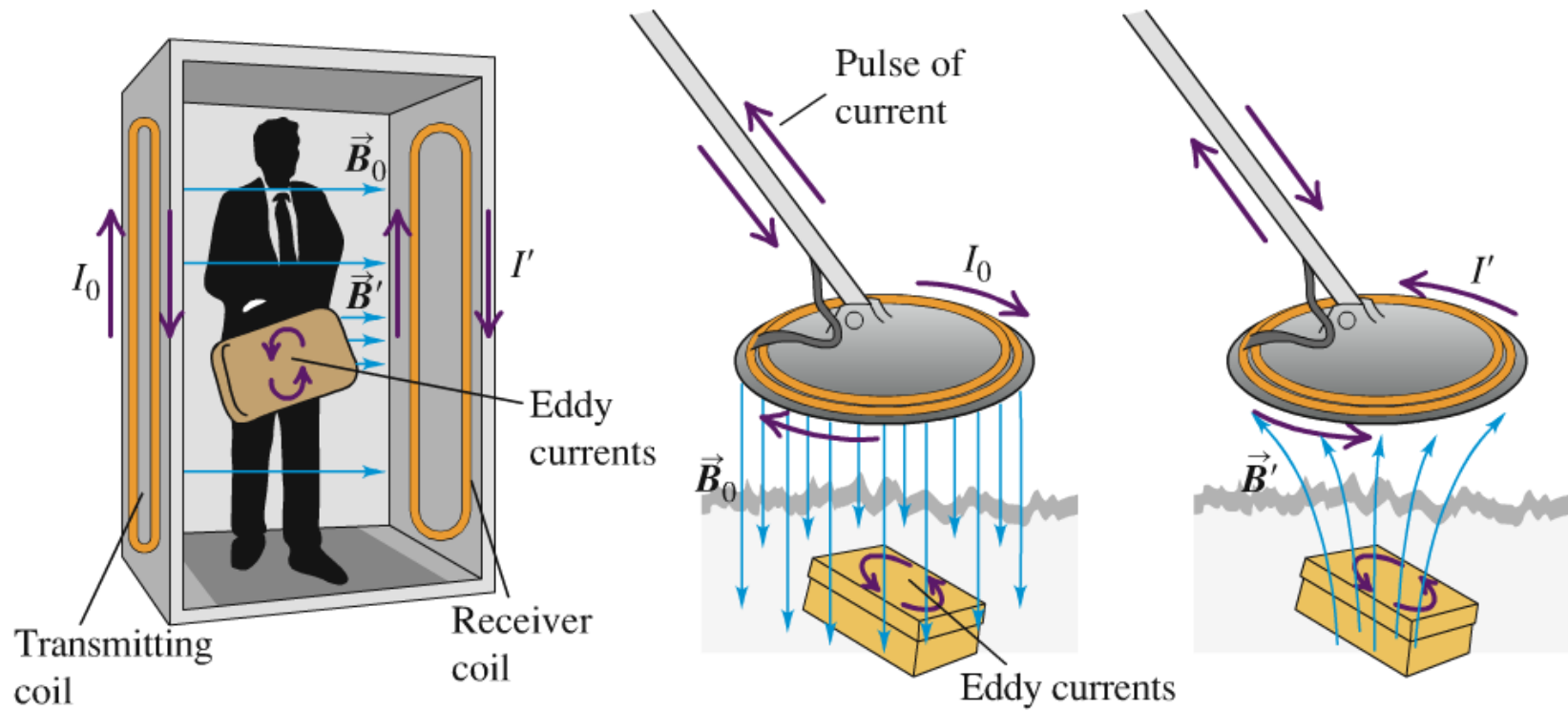


In the configuration of the middle figure, such **eddy current** transfers the kinetic energy of the plate-like pendulum to its thermal energy, so the pendulum will stop after a while.



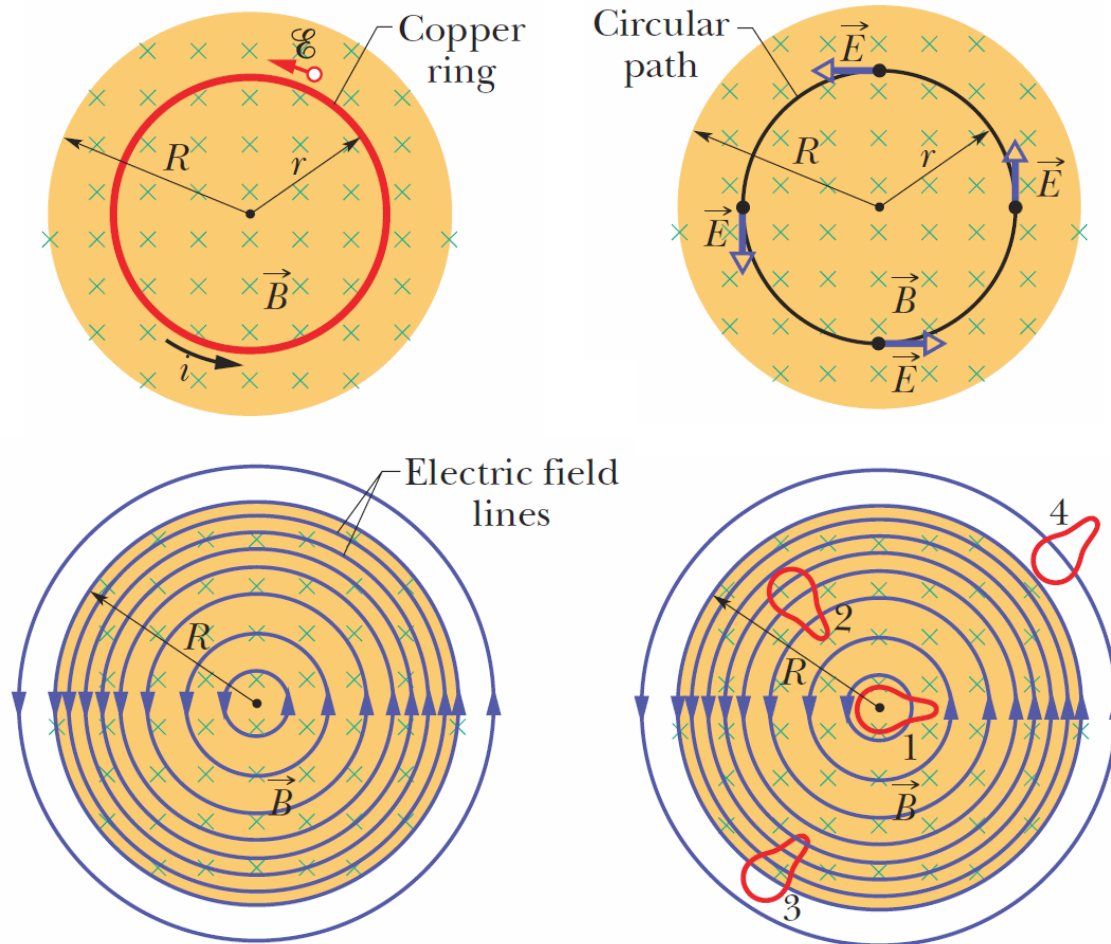
Eddy current braking

Eddy current – metal detector



Induced electric fields

An induced emf exists in a changing **B**-field even if no copper ring is present. A *changing* magnetic field produces an *electric field*.



Faraday's law in terms of electric field

For any closed loop in a changing magnetic field, the induced emf can be calculated as

$$\mathcal{E} = \oint_1 \mathbf{E} \cdot d\mathbf{l}.$$

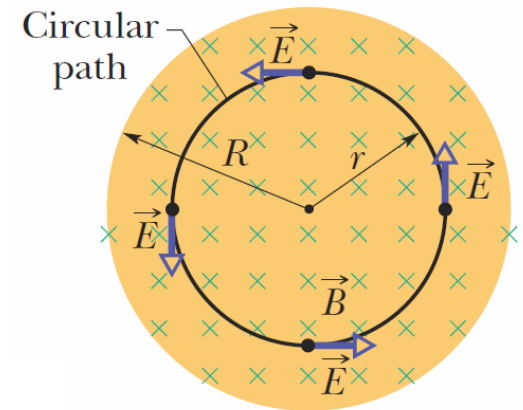
For example, in the case of the right figure,

$$\mathcal{E} = \oint_1 \mathbf{E} \cdot d\mathbf{l} = E \oint_1 d\mathbf{l} = 2\pi r E.$$

With this, the Faraday's law of induction can be rewritten as

$$\oint_1 \mathbf{E} \cdot d\mathbf{l} = - \frac{d\Phi_B}{dt} = - \frac{d}{dt} \int_A \mathbf{B} \cdot d\mathbf{A}.$$

Lenz's law



Faraday's Law & Kirchhoff's loop rule

Kirchhoff's loop rule : The algebraic sum of potential differences around any loop must be zero

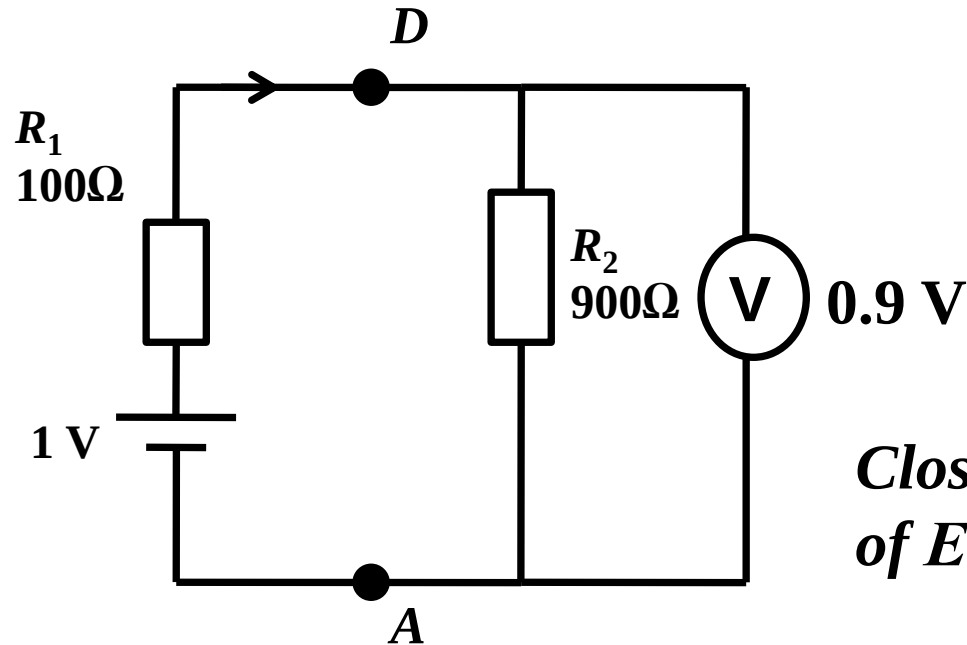
$$\oint \vec{E} \cdot d\vec{s} = 0$$

This rule only holds for a conservative field!
It is not true if we have a changing magnetic flux!

If we have a changing magnetic flux, the E-field inside the wire become non-conservative. The path does matter and **the concept of electric potential does not apply for electric fields produced via induction.**

Faraday's law always holds. Kirchhoff's loop rule is just a special case of Faraday's law when change of flux is zero.

Faraday's Law & Kirchhoff's loop rule



*Closed loop integral
of $E \cdot ds = 0$*

$$I = 0.001A$$

$$V_D - V_A = IR_2 = 0.9V$$

$$V_D - V_A = 1V - IR_1 = 0.9V$$

Faraday's Law & Kirchhoff's loop rule

$$\frac{d\Phi_B}{dt} = 1V$$

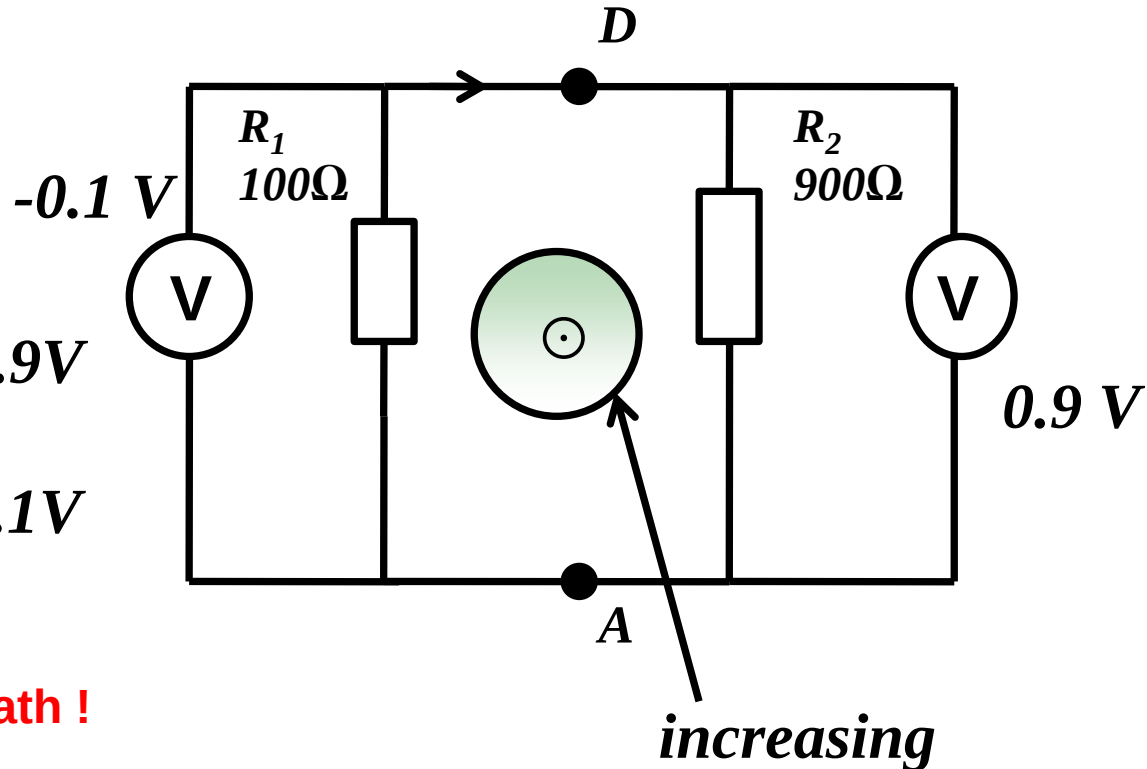
$$I = 0.001A$$

$$\text{Right : } V_D - V_A = IR_2 = +0.9V$$

$$\text{Left : } V_D - V_A = -IR_1 = -0.1V$$

$$\int_A^D \vec{E} \cdot d\vec{s} \quad \text{depends on the path !}$$

Kirchhoff's rule does not hold
Faraday's Law always true



$$\begin{array}{r} V_D - V_A = 0.9V \\ + \\ V_A - V_D = 0.1V \\ \hline V_D - V_D = 1V \end{array}$$

The concept of electric potential does not apply for electric fields produced via induction!

Example

left and right closed loop

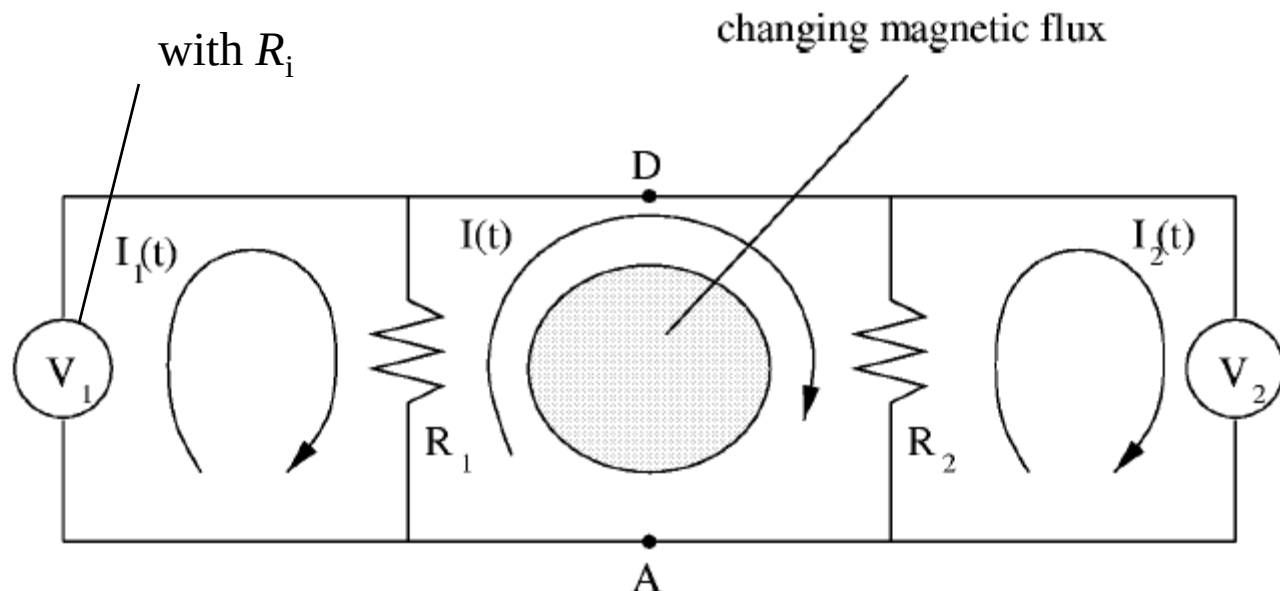
apply Kirchhoff's rule:

$$\oint \vec{E} \cdot d\vec{l} = 0$$

middle closed loop

$$\oint \vec{E} \cdot d\vec{l} = \mathcal{E}(t) = -\frac{d\phi_B}{dt}$$

$\mathcal{E}$ is independent of R_1 and R_2



$$\therefore \begin{aligned} |V_1| &= I_1 R_i \approx IR_1 \\ |V_2| &= I_2 R_i \approx IR_2 \end{aligned}$$

Left loop $I_1 R_i + \cancel{I_1 R_1} - IR_1 = 0$

Middle loop $IR_1 + IR_2 - \cancel{I_1 R_1} - \cancel{I_2 R_2} = \mathcal{E}$

Right loop $\cancel{I_2 R_2} - IR_2 + I_2 R_i = 0$

$R_i \gg R_1$ and $R_i \gg R_2$, therefore $I_1 \ll I$ and $I_2 \ll I$

$$\mathcal{E} = 1 \text{ Volt}$$

$$R_1 = 100 \, \Omega, R_2 = 900 \, \Omega, \text{ and } R_i = 10^7 \, \Omega$$

$$I \approx 1.0 \times 10^{-3} \text{ A}$$

$$|V_1| \approx \frac{1}{10} \text{ Volt, and } |V_2| \approx \frac{9}{10} \text{ Volt}$$

$$I_1 R_i - IR_1 \approx 0$$

$$I (R_1 + R_2) \approx \mathcal{E}$$

$$I_2 R_i - IR_2 \approx 0$$

Question

Can we have electric field lines forming loops?

In what cases?

What does it mean?

The induced E-field is *not* conservative!

The definition of conservative force is that the work done by the force around any closed loop is zero,

$$W = \oint_C \mathbf{F} \cdot d\mathbf{r} = 0.$$

Apparently, the E-field induced by a changing B-field does not fulfill this condition, since

$$\mathcal{E} = \oint_1 \mathbf{E} \cdot d\mathbf{l} \neq 0.$$

Therefore, **we cannot define a scalar potential U in the induced E-field**. This is similar to the case of the B-field produced by moving charges (currents) where

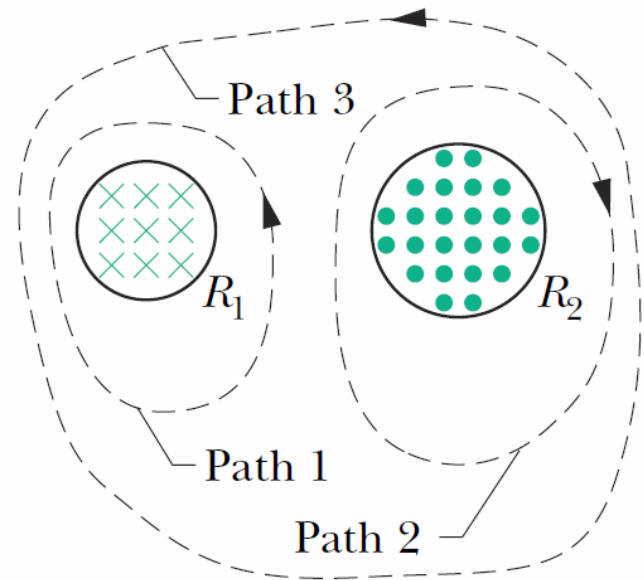
$$\oint_1 \mathbf{B} \cdot d\mathbf{l} = \mu_0 \int_{\mathbf{A}} \mathbf{J} \cdot d\mathbf{A} \neq 0.$$

Example

The right figure shows two circular regions R_1 and R_2 with radii r_1 and r_2 ($r_2 > r_1$). In R_1 there is a uniform magnetic field of magnitude B_1 directed into the page, and in R_2 there is a uniform magnetic field of magnitude B_2 directed out of the page (ignore fringing). Both fields are decreasing at the rate of δ . Calculate $\oint \mathbf{E} \cdot d\mathbf{l}$ for (a) Path 1, (b) Path 2 and (c) Path 3.

Solve:

(a) In R_1 , the field into the page is decreasing. If we assume that there was a wire along Path 1, then the B-field produced by the wire would be pointing into the page. Therefore, the “current” should be clockwise.



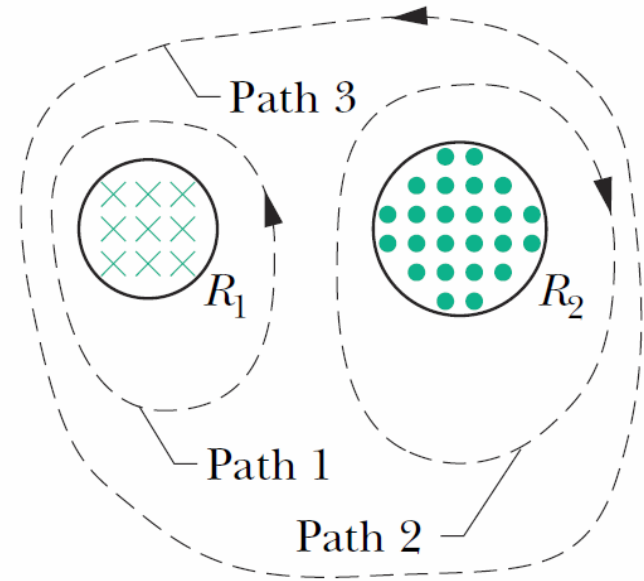
Example

(a) So, $\oint \mathbf{E} \cdot d\mathbf{l}$ is clockwise, which is opposite to the direction of Path 1. Therefore, $\oint \mathbf{E} \cdot d\mathbf{l}$ along Path 1 should be negative.

$$\oint_{P_1} \mathbf{E} \cdot d\mathbf{l} = -A_{R_1} \frac{dB}{dt} = -\pi r_1^2 \delta.$$

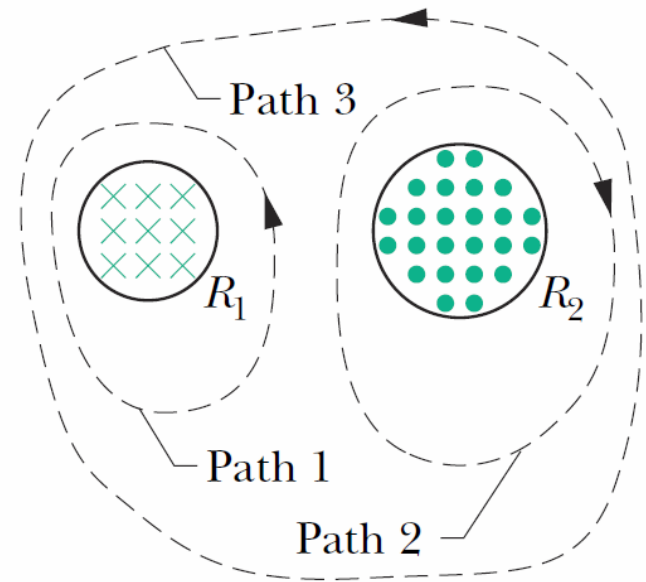
(b) Similar to (a), we conclude that $\oint \mathbf{E} \cdot d\mathbf{l}$ along Path 2 should also be negative.

$$\oint_{P_2} \mathbf{E} \cdot d\mathbf{l} = -A_{R_2} \frac{dB}{dt} = -\pi r_2^2 \delta.$$



Example

(c) Since $\oint \mathbf{E} \cdot d\mathbf{l}$ for R_1 (not Path 1!) is clockwise, and that for R_2 is counterclockwise, the net $\oint \mathbf{E} \cdot d\mathbf{l}$ for a loop enclosing R_1 and R_2 should be counterclockwise (because $r_2 > r_1$). Therefore, $\oint \mathbf{E} \cdot d\mathbf{l}$ along R_3 should be positive.



$$\oint_{P_3} \mathbf{E} \cdot d\mathbf{l} = A_{R_2} \frac{dB}{dt} - A_{R_1} \frac{dB}{dt} = \pi \delta (r_2^2 - r_1^2).$$

The Differential Form of Faraday's Law*

The integral form of the Faraday's law: $\oint_1 \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int_{\mathbf{A}} \mathbf{B} \cdot d\mathbf{A}$
Introduce again the Nabla symbol

$$\nabla \equiv \frac{\partial}{\partial x} \mathbf{e}_x + \frac{\partial}{\partial y} \mathbf{e}_y + \frac{\partial}{\partial z} \mathbf{e}_z$$

The mathematical **curl theorem (Stokes' theorem)** relates the surface integral of the *curl* of a vector field $\mathbf{F}$ over a surface $\mathbf{A}$ in Euclidean three-space to the line integral of the vector field over its boundary $\partial\mathbf{A}$:

$$\int_{\mathbf{A}} \nabla \times \mathbf{F} \cdot d\mathbf{A} = \oint_{1=\partial\mathbf{A}} \mathbf{F} \cdot d\mathbf{l}$$

Apply this theorem to the left of the Faraday's law, we get

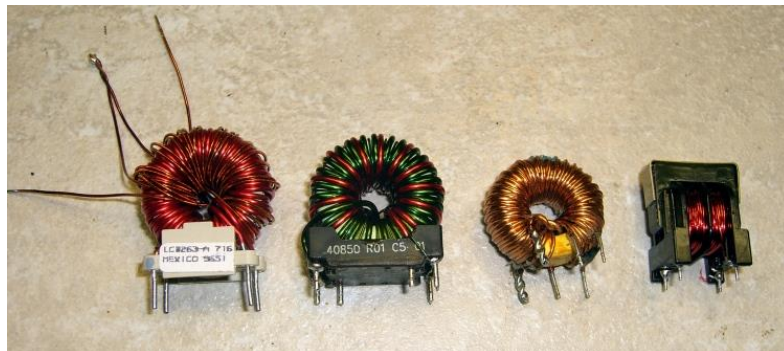
$$\oint_{1=\partial\mathbf{A}} \mathbf{E} \cdot d\mathbf{l} = \int_{\mathbf{A}} \nabla \times \mathbf{E} \cdot d\mathbf{A} = -\frac{d}{dt} \int_{\mathbf{A}} \mathbf{B} \cdot d\mathbf{A}.$$

Getting rid of the integral yields the differential form of the Faraday's law:

$$\nabla \times \mathbf{E} = -\frac{d\mathbf{B}}{dt}$$

Applications: Inductors

A solenoid can be viewed as the simplest form of an **inductor** who produces a magnetic flux in a circuit.



INDUCTOR COLOUR CODE

Result: μH

4-BAND-CODE $\rightarrow$ $270\mu\text{H} \pm 5\%$

COLOR	1st BAND	2nd BAND	MULTIPLIER	TOLERANCE
BLACK	0	0	1	$\pm 20\%$
BROWN	1	1	10	$\pm 1\%$
RED	2	2	100	$\pm 2\%$
ORANGE	3	3	1,000	$\pm 3\%$
YELLOW	4	4	10,000	$\pm 4\%$
GREEN	5	5		
BLUE	6	6		
VIOLET	7	7		
GREY	8	8		
WHITE	9	9		
GOLD			0.1	5%
SILVER			0.01	10%

Result: μH

$27\mu\text{H} \pm 5\%$

GOLD			decimal point	5%
------	--	--	---------------	----

Inductance

If a solenoid inductor has N turns, and produces a magnetic flux Φ_B when the wire has current i , then its inductance is defined as

$$L = \frac{N\Phi_B}{i}.$$

Note that, similar to the resistance and the capacitance, the above formula is only used to *calculate* the inductance. The inductance of an inductor is *determined* by its geometrical factors.

The unit of inductance is the henry (H):

$$1 \text{ henry} = 1 \text{ H} = 1 \text{ T} \cdot \text{m}^2/\text{A}.$$

Inductance of a solenoid

From Chapter 29 we knew that, if a solenoid inductor of length l has n turns per unit length and its cross section area is A , then the field produced by current i is

$$B = \mu_0 i n.$$

So, with the definition of conductance we have

$$L = \frac{N\Phi_B}{i} = \frac{(nl)(\mu_0 i n A)}{i} = \mu_0 n^2 l A.$$

Therefore the conductance of a solenoid per unit length is

$$\frac{L}{l} = \mu_0 n^2 A.$$

Using unit analysis on the above equation, we can get the other unit of the permeability constant μ_0 : $[\mu_0] = \text{H/m}$.

Inductance of a rounded solenoid

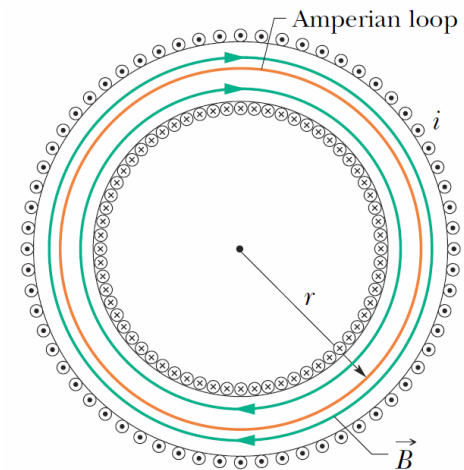
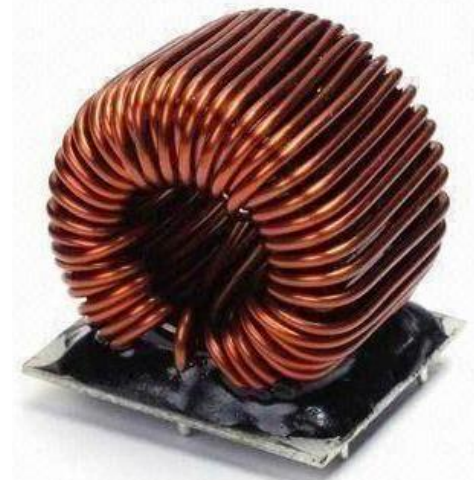
Many inductors have the shape of a rounded solenoid (right figure). Let the outer and inner radii of the torus be r_2 and r_1 , the width of the solenoid be b , the total number of turns be N , from Chapter 29 we knew that B is only a function of r :

$$B = \frac{\mu_0 N i}{2\pi r}.$$

Then the cross-sectional magnetic flux is

$$\Phi_B = \int_A B dA = \frac{\mu_0 N i b}{2\pi} \int_{r_1}^{r_2} \frac{dr}{r} = \frac{\mu_0 N i b}{2\pi} \ln \frac{r_2}{r_1}.$$

So the inductance is $L = \frac{\mu_0 N^2 b}{2\pi} \ln \frac{r_2}{r_1}.$



Ch30 Key Points

1. Faraday's law and Lenz's law
2. Induction and energy transfers
3. Induced electric fields
4. Inductors and Inductance
5. Self-induction
6. *RL* circuits
7. Energy stored in a magnetic field
8. Energy density of a magnetic field
9. Mutual induction

Application of inductors in fluorescent tube

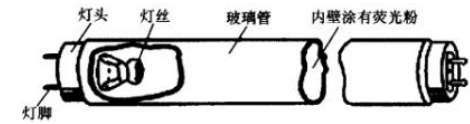
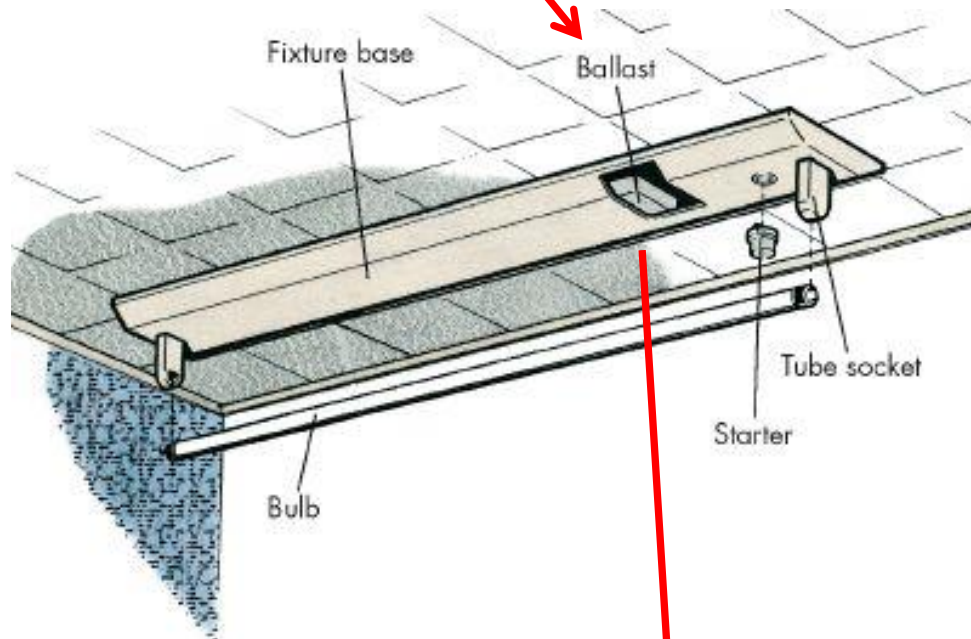


图 5-69 日光灯结构

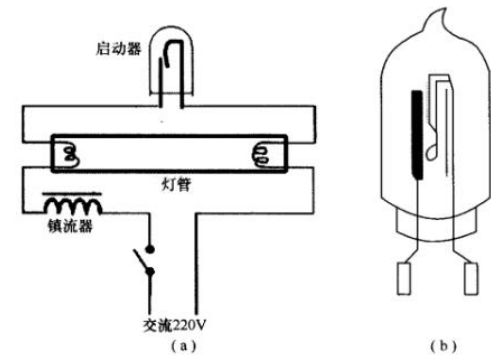
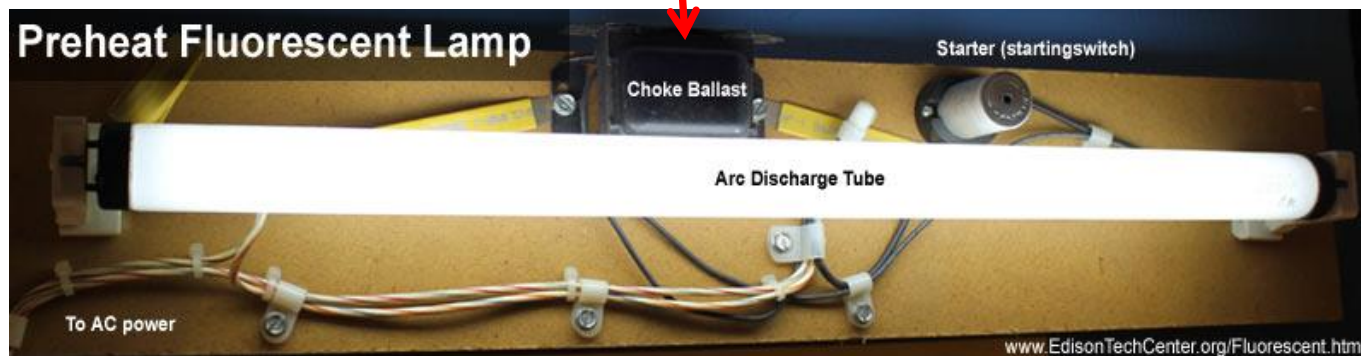
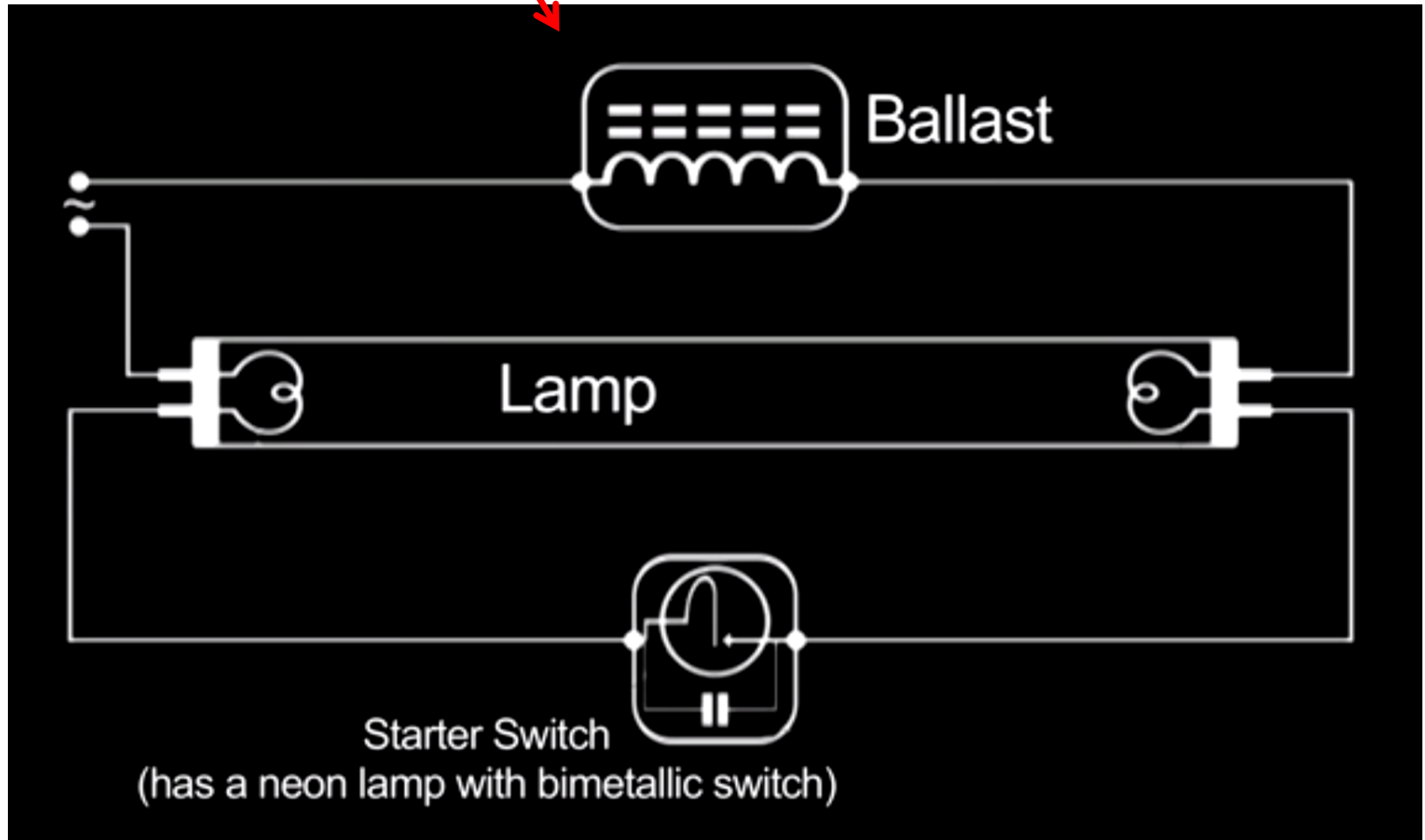


图 5-70 日光灯电路图

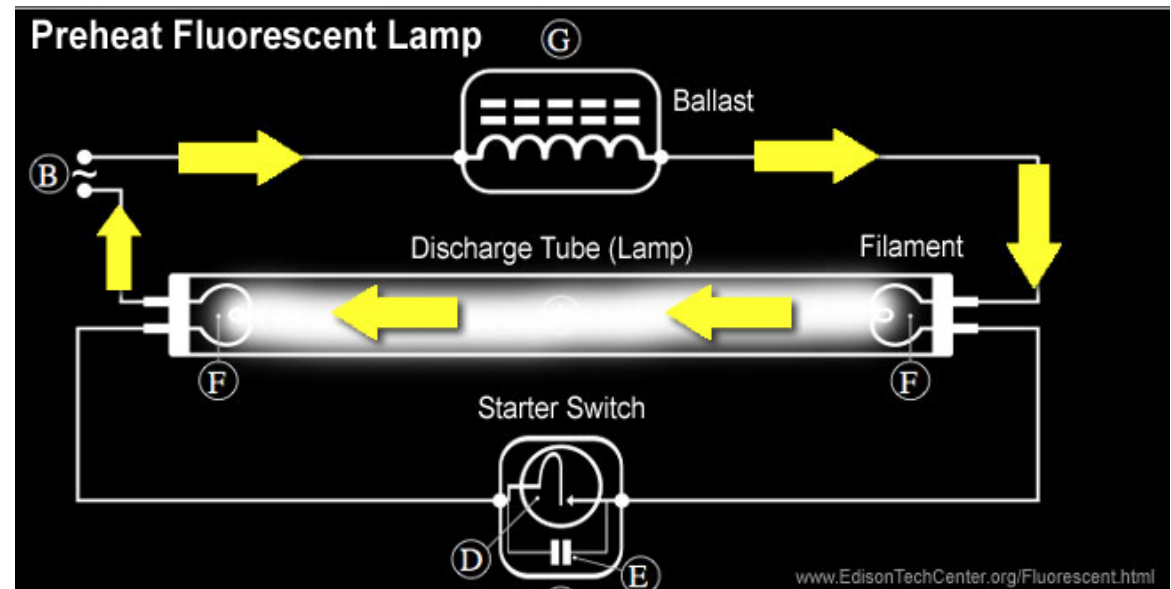
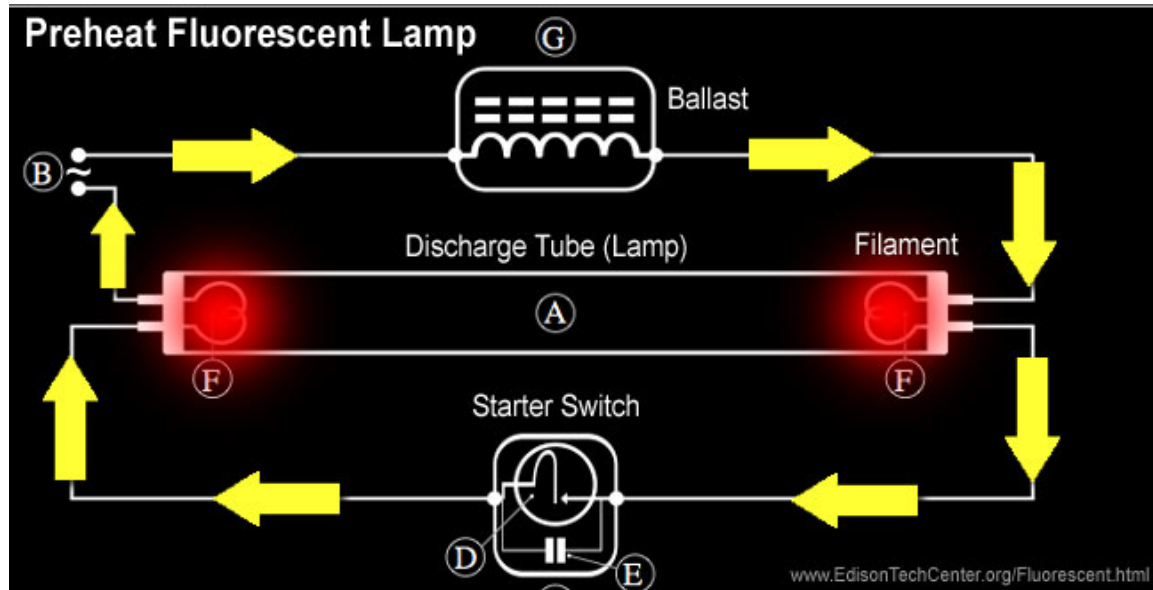
(a) 日光灯电路; (b) 启动器。



Application of inductors in fluorescent tube



Application of inductors in fluorescent tube





Self induction

Due to Faraday's law, if the current of a solenoid inductor is *changing*, there must be an induced emf

$$\mathcal{E}_L = -\frac{d(N\Phi_B)}{dt}.$$

Combined with $L = N\Phi_B/i$, we have

$$\mathcal{E}_L = -L \frac{di}{dt}.$$

This is called the *self-induced* emf.

Direction: The minus sign indicates that the direction of the self-induced emf oppose the change of current i (Lenz's law).

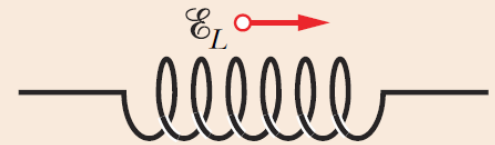
Potential: Although electric potential V cannot be defined in the inductor L , it *can* be defined at any point out of L .

Self induction



CHECKPOINT 5

The figure shows an emf $\mathcal{E}_L$ induced in a coil. Which of the following can describe the current through the coil: (a) constant and rightward, (b) constant and leftward, (c) increasing and rightward, (d) decreasing and rightward, (e) increasing and leftward, (f) decreasing and leftward?



57 At a given instant the current and self-induced emf in an inductor are directed as indicated in Fig. 30-47.

(a) Is the current increasing or decreasing? (b) The induced emf is 23 V, and the rate of change of the current is 18 kA/s; find the inductance.

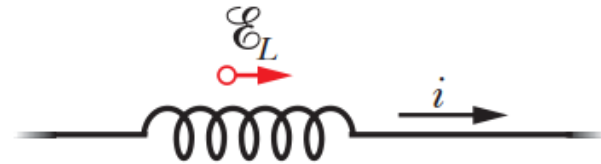


Figure 30-47 Problem 57.

RL circuits

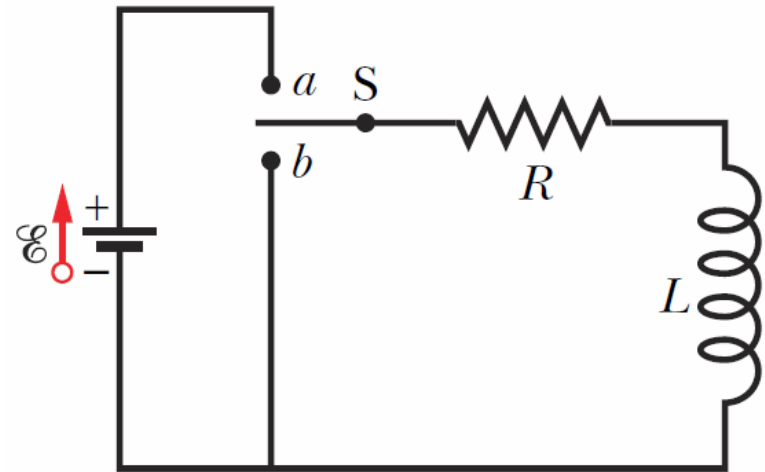
An RL circuit consists of a resistor R and an inductor L . When the switch is closed on a , the current starts to increase, which produces a self-induced emf $\mathcal{E}_L$ opposing the battery emf $\mathcal{E}$. With time, $\mathcal{E}_L$ decreases and goes to zero, i approaches $\mathcal{E}/R$. Then the switch is closed on b to release the energy stored.

Current increasing: The current and the voltage across L are functions of time,

$$i = i(t), \quad \mathcal{E}_L = -L \frac{di}{dt} = \mathcal{E}_L(t).$$

We use Kirchhoff's voltage law to find relation between them:

$$\mathcal{E} - i(t)R - L \frac{di}{dt} = 0.$$



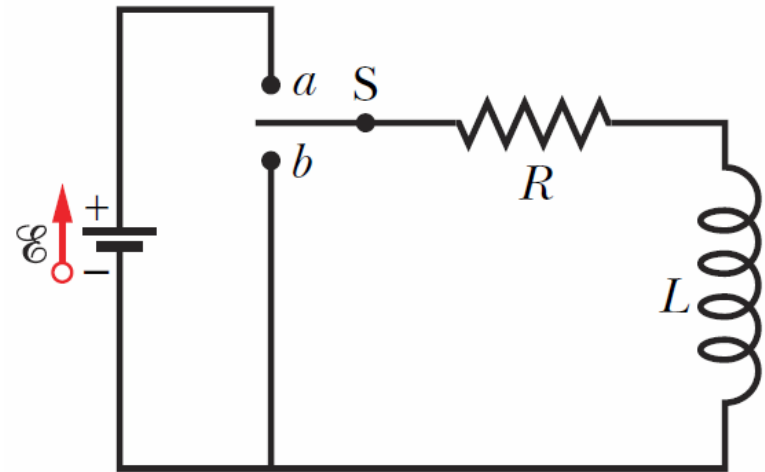
RL circuits

Thus we obtain the governing differential equation for the current-increasing process:

$$\frac{di}{dt} + \frac{iR}{L} = \frac{\mathcal{E}}{L}.$$

This differential equation, plus the condition that current is zero at time zero (called a *boundary condition*), completes the mathematical problem:

$$\frac{di}{dt} + \frac{iR}{L} = \frac{\mathcal{E}}{L}, \quad i(0) = 0.$$



The solution of this problem, $i = i(t)$, consists of a general solution $i_0(t)$ that satisfy the reduced equation **in which the right side is zero**, plus a *particular solution* that satisfies the entire equation:

$$i = i(t) = i_0(t) + i_p.$$

Since the derivative of an exponential function has basically the same form of itself, $d(e^{bt})/dt = be^{bt}$, we assume that the general solution has the form

$$i_0(t) = Ae^{bt}.$$

Plug this form into the equation, we have

$$Abe^{bt} + \frac{Ae^{bt}R}{L} = 0, \quad \text{i.e., } b = -\frac{R}{L}.$$

RL circuits

We define $\tau_L = L/R$ the *time constant of a RL circuit*.

The next task is to find the particular solution to the equation.
If i is a constant, $di/dt = 0$, the equation becomes

$$0 + \frac{i_p R}{L} = \frac{\mathcal{E}}{L}, \quad \text{so} \quad i_p = \frac{\mathcal{E}}{R}.$$

So far we have

$$i(t) = i_0(t) + i_p = Ae^{-tR/L} + \frac{\mathcal{E}}{R},$$

We finally use $i(0) = 0$ to determine A :

$$\left[Ae^{-tR/L} + \frac{\mathcal{E}}{R} \right]_{t=0} = A + \frac{\mathcal{E}}{R} = 0, \quad A = -\frac{\mathcal{E}}{R}.$$

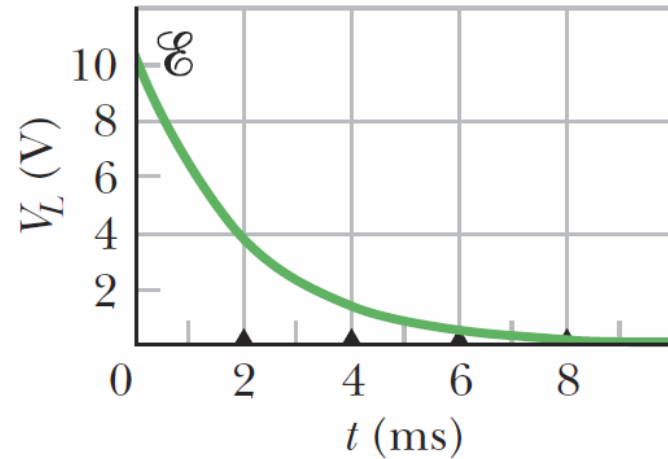
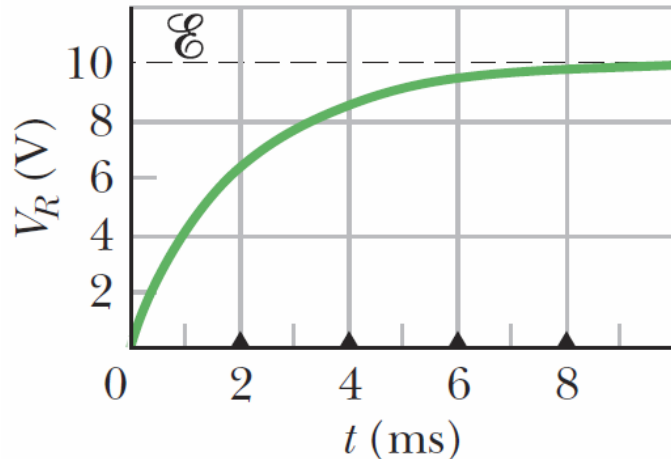
RL circuits

Therefore, the final solution of the original differential equation is

$$i(t) = \frac{\mathcal{E}}{R}(1 - e^{-t/\tau_L})$$

The potential difference across L is then

$$\mathcal{E}_L(t) = -L \frac{di}{dt} = -\mathcal{E} e^{-t/\tau_L}.$$



Current decreasing: The governing differential equation is similar to that of the current increasing process, but since there's no battery, the right side is zero. The boundary condition is now $i(0) = i_0 = \mathcal{E}/R$.

$$\frac{di}{dt} + \frac{iR}{L} = 0, \quad i(0) = i_0.$$

The general solution of this equation is just

$$i(t) = Ae^{-tR/L}.$$

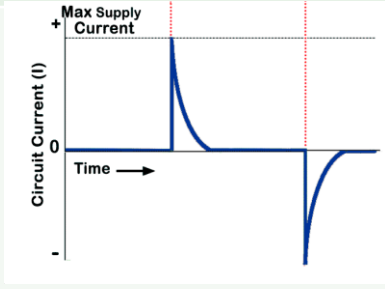
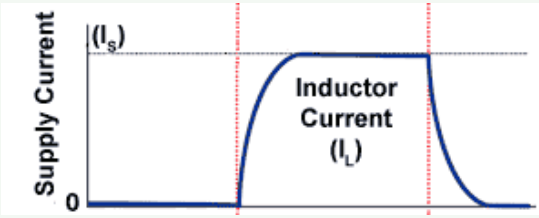
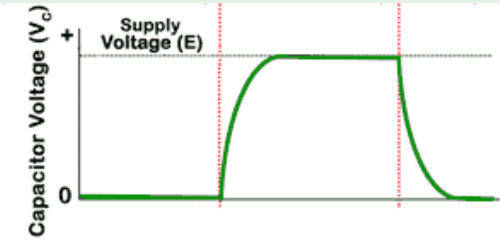
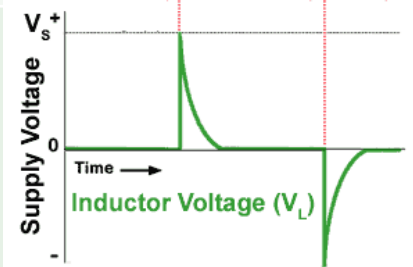
Using $i(0) = i_0$, we obtain the final solution:

$$i(t) = i_0 e^{-tR/L}.$$

And

$$\mathcal{E}_L(t) = -L \frac{di}{dt} = i_0 R e^{-tR/L} = \mathcal{E} e^{-tR/L}.$$

Comparison between *RC* and *RL* circuits

	RC Circuit	RL Circuit
Governing equation	$\frac{dq}{dt} + \frac{q}{RC} = \frac{\mathcal{E}}{R}$	$\frac{di}{dt} + \frac{iR}{L} = \frac{\mathcal{E}}{L}$
Time constant	$\tau_C = RC$	$\tau_L = R/L$
Solution (q/i increasing)	$q(t) = C\mathcal{E}(1 - e^{-t/\tau_C})$	$i(t) = \frac{\mathcal{E}}{R}(1 - e^{-t/\tau_L})$
<i>i-t</i> curve		
<i>V-t</i> curve		

Energy stored in a magnetic field

Consider again the governing equation of the RL circuit in the case of current increasing, multiplying both sides by i :

$$i\mathcal{E} = Li \frac{di}{dt} + i^2 R$$

Left term: Power of the battery – energy delivered from the battery to the circuit per unit time.

Rightmost term: Power of the resistor.

First term on the right side: Power of the inductor – energy stored in the inductor's *magnetic field* per unit time.

$$\frac{dU_B}{dt} = Li \frac{di}{dt}$$

Integration on both sides gives the energy in the B-field:

$$U_B = \int_0^t Li di = \frac{1}{2} Li^2.$$

Energy density of a magnetic field

Similar to our treatment for the energy density of the E-field, we consider the uniform B-field within a solenoid (cross section A , length l). Its energy density is

$$u_B = \frac{U_B}{Al} = \frac{Li^2}{2Al}.$$

Since $L/l = \mu_0 n^2 A$ and $B = \mu_0 i n$ for a solenoid, we have

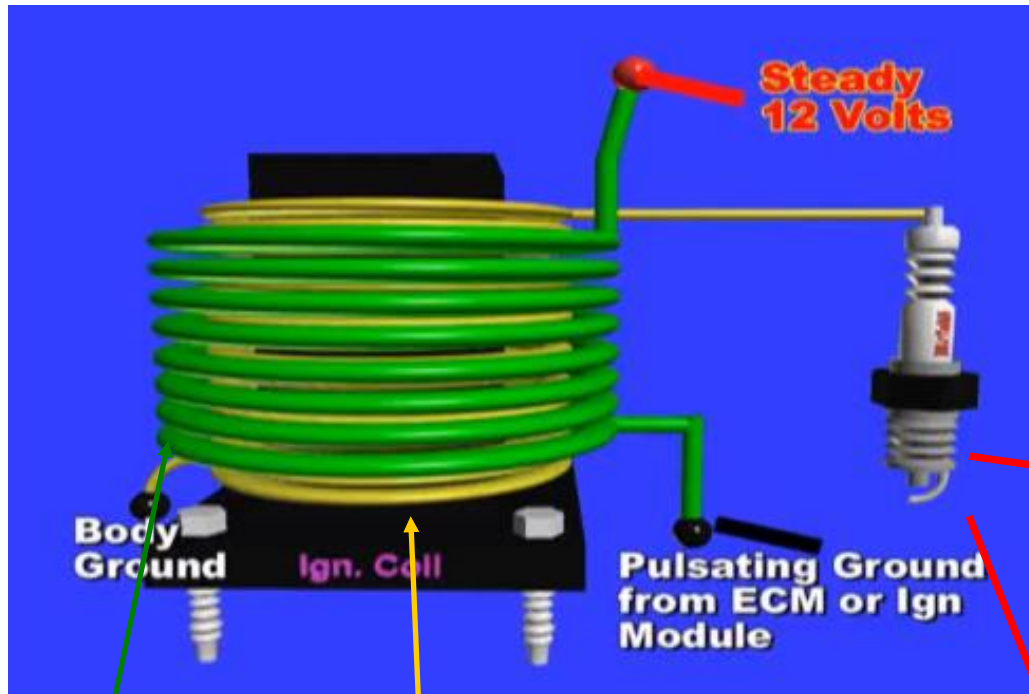
$$u_B = \frac{Li^2}{2Al} = \frac{\mu_0 n^2 A i^2}{2A} = \frac{\mu_0 n^2 i^2}{2} = \frac{B^2}{2\mu_0}.$$

Similar to the energy density in the E-field, $u_E = \frac{1}{2} \epsilon_0 E^2$, the expression for u_B can be used in any B-field.

Expressions of energy we learned

	Energy	Energy density
Kinetic energy	$E_k = \frac{1}{2}mv^2$	/
Vibrational energy	$E_v = \frac{1}{2}kA^2$	/
Rotational energy	$E_r = \frac{1}{2}I\omega^2$	/
Energy in E-field	$U_E = \frac{1}{2}CV^2$	$u_E = \frac{1}{2}\epsilon_0 E^2$
Energy in B-field	$U_B = \frac{1}{2}Li^2$	$u_B = \frac{1}{2\mu_0} B^2$

Example : Magnetic field energy in the ignition system of an engine



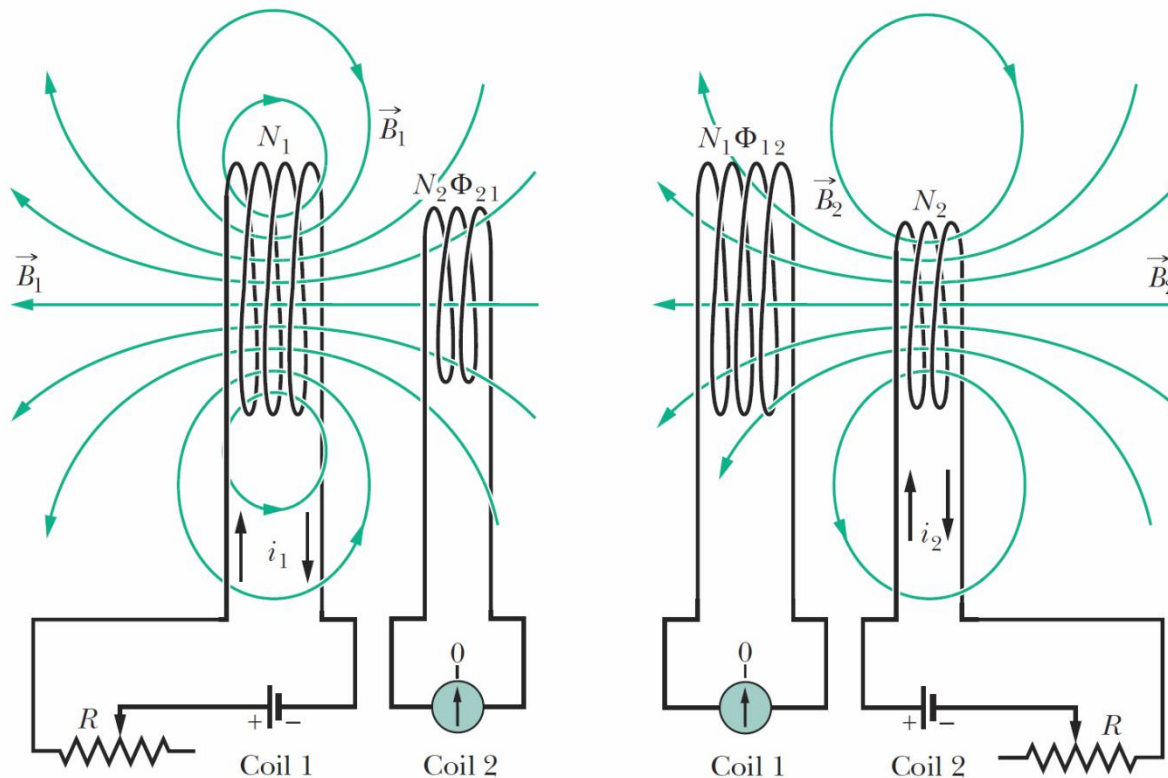
~250 turns

~25,000 turns



Mutual induction

In a system with two close-by coils, if we change the current in one coil, the Φ_B in the other coil will also change, triggering induction current in the latter. This is **mutual induction**.



Mutual induction

If we mark the magnetic flux in Coil 2 (N_2 turns) due to the current i_1 in Coil 1 as Φ_{21} , then the **mutual inductance** of Coil 2 with respect to Coil 1 is

$$M_{21} = \frac{N_2 \Phi_{21}}{i_1}.$$

Multiplying both sides by i_1 and take the derivative w.r.t. t ,

$$M_{21} \frac{di_1}{dt} = N_2 \frac{d\Phi_{21}}{dt}.$$

Now the right side is just the minus of $\mathcal{E}_2$, so

$$\mathcal{E}_2 = -M_{21} \frac{di_1}{dt}.$$

Using the same logic, we have $\mathcal{E}_1 = -M_{12} \frac{di_2}{dt}$.

Mutual induction

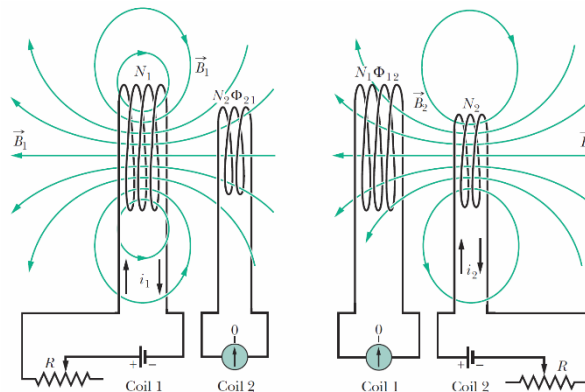
We state here without prove that

$$M \equiv M_{12} = M_{21}.$$

Therefore,

$$\mathcal{E}_2 = -M \frac{di_1}{dt}, \quad \mathcal{E}_1 = -M \frac{di_2}{dt}.$$

Think: What are the rules for inductors in series and parallel??
(Hint: Use $\mathcal{E} = -L di/dt$)

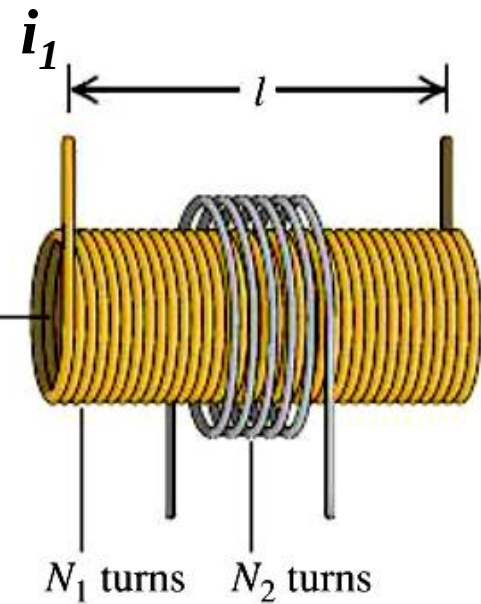


Example – finding the mutual inductance

$$\begin{aligned} M &= \frac{N_2 \Phi_{B2}}{i_1} = \frac{N_2 B_1 A}{i_1} \\ &= \frac{N_2}{i_1} \frac{\mu_0 N_1 i_1}{l} A \\ &= \frac{\mu_0 A N_1 N_2}{l} \end{aligned}$$

$$\begin{aligned} B_1 &= \mu_0 n_1 i_1 \\ &= \frac{\mu_0 N_1 i_1}{l} \end{aligned}$$

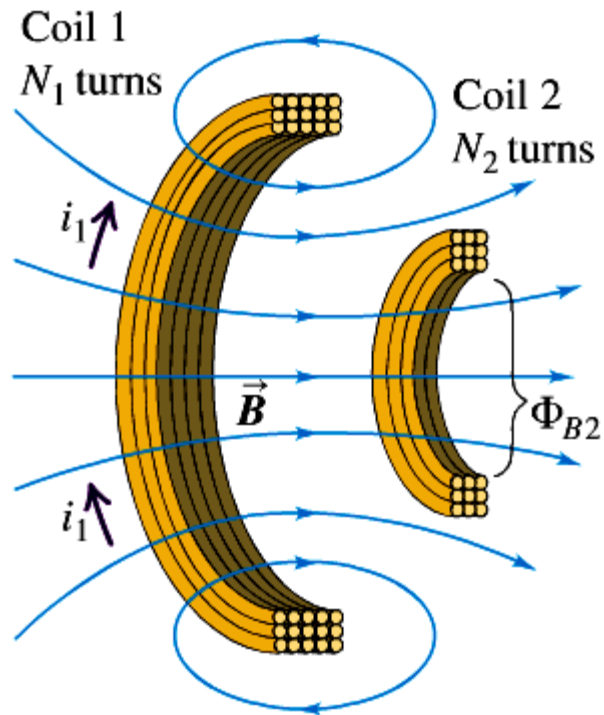
Cross-sectional area A



Depend only on the geometry!

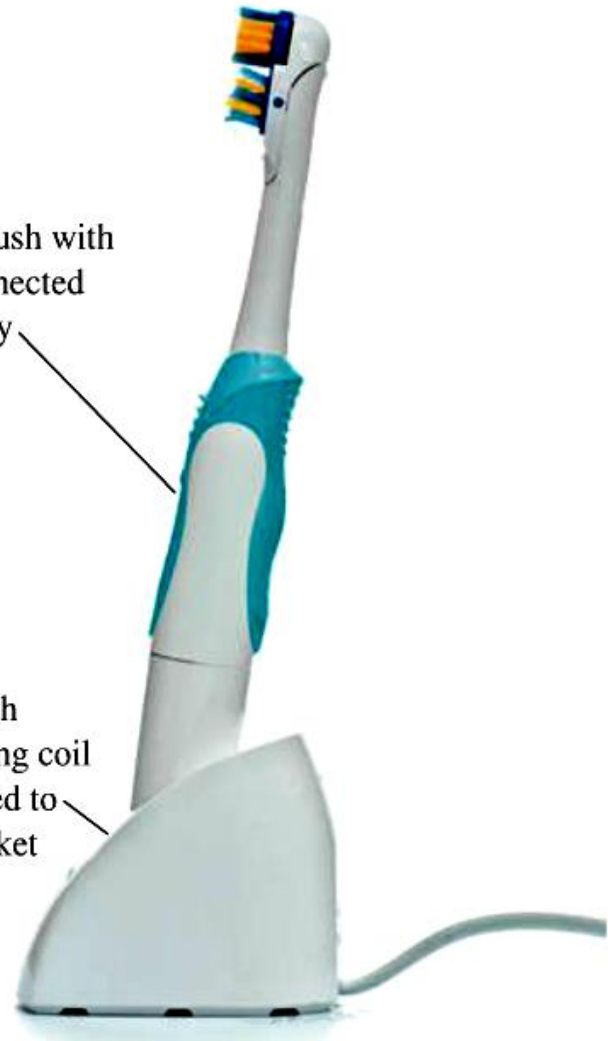
If I double the number of turns in coil 1 and double the length l , how will M change?

Mutual inductance: If the current in coil 1 is changing, the changing flux through coil 2 induces an emf in coil 2.



Toothbrush with
coil connected
to battery

Base with
recharging coil
connected to
wall socket



Magnetic Flux The *magnetic flux* Φ_B through an area A in a magnetic field $\vec{B}$ is defined as

$$\Phi_B = \int \vec{B} \cdot d\vec{A}, \quad (30-1)$$

where the integral is taken over the area. The SI unit of magnetic flux is the **weber**, where $1 \text{ Wb} = 1 \text{ T} \cdot \text{m}^2$. If $\vec{B}$ is perpendicular to the area and uniform over it, Eq. 30-1 becomes

$$\Phi_B = BA \quad (\vec{B} \perp A, \vec{B} \text{ uniform}). \quad (30-2)$$

Faraday's Law of Induction If the magnetic flux Φ_B through an area bounded by a closed conducting loop changes with time, a current and an emf are produced in the loop; this process is called *induction*. The induced emf is

$$\mathcal{E} = - \frac{d\Phi_B}{dt} \quad (\text{Faraday's law}). \quad (30-4)$$

If the loop is replaced by a closely packed coil of N turns, the induced emf is

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}. \quad (30-5)$$

Lenz's Law An induced current has a direction such that the magnetic field *due to the current* opposes the change in the magnetic flux that induces the current. The induced emf has the same direction as the induced current.

Emf and the Induced Electric Field An emf is induced by a changing magnetic flux even if the loop through which the flux is changing is not a physical conductor but an imaginary line. The changing magnetic field induces an electric field $\vec{E}$ at every point of such a loop; the induced emf is related to $\vec{E}$ by

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{s}, \quad (30-19)$$

where the integration is taken around the loop. From Eq. 30-19 we can write Faraday's law in its most general form,

$$\oint \vec{E} \cdot d\vec{s} = - \frac{d\Phi_B}{dt} \quad (\text{Faraday's law}). \quad (30-20)$$

A changing magnetic field induces an electric field $\vec{E}$.

Inductors An **inductor** is a device that can be used to produce a known magnetic field in a specified region. If a current i is established through each of the N windings of an inductor, a magnetic flux Φ_B links those windings. The **inductance** L of the inductor is

$$L = \frac{N\Phi_B}{i} \quad (\text{inductance defined}). \quad (30-28)$$

The SI unit of inductance is the **henry** (H), where $1 \text{ henry} = 1 \text{ H} = 1 \text{ T} \cdot \text{m}^2/\text{A}$. The inductance per unit length near the middle of a long solenoid of cross-sectional area A and n turns per unit length is

$$\frac{L}{l} = \mu_0 n^2 A \quad (\text{solenoid}). \quad (30-31)$$

Self-Induction If a current i in a coil changes with time, an emf is induced in the coil. This self-induced emf is

$$\mathcal{E}_L = -L \frac{di}{dt}. \quad (30-35)$$

The direction of $\mathcal{E}_L$ is found from Lenz's law: The self-induced emf acts to oppose the change that produces it.

Series RL Circuits If a constant emf $\mathcal{E}$ is introduced into a single-loop circuit containing a resistance R and an inductance L , the current rises to an equilibrium value of $\mathcal{E}/R$:

$$i = \frac{\mathcal{E}}{R} (1 - e^{-t/\tau_L}) \quad (\text{rise of current}). \quad (30-41)$$

Here $\tau_L (= L/R)$ is the **inductive time constant**. When the source of constant emf is removed, the current decays from a value i_0 according to

$$i = i_0 e^{-t/\tau_L} \quad (\text{decay of current}). \quad (30-45)$$

Magnetic Energy If an inductor L carries a current i , the inductor's magnetic field stores an energy given by

$$U_B = \frac{1}{2} Li^2 \quad (\text{magnetic energy}). \quad (30-49)$$

If B is the magnitude of a magnetic field at any point (in an inductor or anywhere else), the density of stored magnetic energy at that point is

$$u_B = \frac{B^2}{2\mu_0} \quad (\text{magnetic energy density}). \quad (30-55)$$

Mutual Induction If coils 1 and 2 are near each other, a changing current in either coil can induce an emf in the other. This mutual induction is described by

$$\mathcal{E}_2 = -M \frac{di_1}{dt} \quad (30-64)$$

and

$$\mathcal{E}_1 = -M \frac{di_2}{dt}, \quad (30-65)$$

where M (measured in henries) is the mutual inductance.

Comparison between *E* and *M* (1)

	Electricity		Magnetism	
Physical Quantities	constant	$\frac{1}{4\pi\epsilon_0}$	constant	$\frac{\mu_0}{4\pi}$
	charge	q	current	i
	E-field	$\vec{E}$	B-field	$\vec{B}$
	E-potential	V	B-potential	To be continued
	capacity	C	inductance	L
	E-dipole	$\vec{p}$	B-dipole	$\vec{\mu}$

E means electric field; B means magnetic field

Comparison between E and M (2)

	Electricity		Magnetism	
Force	$\vec{F} = q\vec{E}$		$\vec{F} = q(\vec{v} \times \vec{B})$	
Field	Coulomb	$d\vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2} \hat{r}$	Biot-Savart	$d\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{id\vec{s} \times \hat{r}}{r^2}$
	E-Gauss <i>Maxwell 1</i>	$\oiint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$	B-Gauss <i>Maxwell 2</i>	$\oiint \vec{B} \cdot d\vec{A} = 0$
	Faraday <i>Maxwell 3</i>	$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{A}$	Ampere <i>Maxwell 4</i>	$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$
	E-Gauss <i>infinite rod</i>	$E = \frac{\lambda}{2\pi\epsilon_0 r}$	B-Gauss <i>infinite wire</i>	$B = \frac{\mu_0 i}{2\pi r}$
	E-Gauss <i>infinite plane</i>	$E = \frac{\sigma}{2\epsilon_0}$	B-Gauss <i>infinite sheet</i>	$B = \frac{\mu_0 J}{2}$

Comparison between E and M (3)

	Electricity		Magnetism	
Field and Potential	$\Delta V = - \int \vec{E} \cdot d\vec{s}$		To be continued	
Energy	capacitor	$C = \frac{q}{V}$	inductor	$L = \frac{N\Phi_B}{i}$
	capacitor <i>parallel plate</i>	$C = \frac{\epsilon_0 A}{d}$	Inductor <i>solenoid</i>	$L = \mu_0 n^2 l A$
	E-energy	$U_E = \frac{Q^2}{2C}$	B-energy	$U_B = \frac{LI^2}{2}$
	E-energy density	$\mu_E = \frac{1}{2} \epsilon_0 E^2$	B-energy density	$\mu_B = \frac{1}{2\mu_0} B^2$

Comparison between E and M (4)

	Electricity		Magnetism	
Dipole	E-dipole	$\vec{p} = q\vec{d}$	B-dipole	$\vec{\mu} = i\vec{A}$
	torque	$\vec{\tau} = \vec{p} \times \vec{E}$	torque	$\vec{\tau} = \vec{\mu} \times \vec{B}$
	potential energy	$U = -\vec{p} \cdot \vec{E}$	potential energy	$U = -\vec{\mu} \cdot \vec{B}$
	E-field <i>large z</i>	$\vec{E} = \frac{1}{2\pi\epsilon_0} \cdot \frac{\vec{p}}{z^3}$	B-energy density	$\vec{B} = \frac{\mu_0}{2\pi} \cdot \frac{\vec{\mu}}{z^3}$